

MAGMA: A Multi-Graph based Agentic Memory Architecture for AI Agents

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Abstract

Memory-Augmented Generation (MAG) extends Large Language Models with external memory to support long-context reasoning, but existing approaches largely rely on semantic similarity over monolithic memory stores, entangling temporal, causal, and entity information. This design limits interpretability and alignment between query intent and retrieved evidence, leading to suboptimal reasoning accuracy. In this paper, we propose MAGMA, a multi-graph agentic memory architecture that represents each memory item across orthogonal semantic, temporal, causal, and entity graphs. MAGMA formulates retrieval as policy-guided traversal over these relational views, enabling query-adaptive selection and structured context construction. By decoupling memory representation from retrieval logic, MAGMA provides transparent reasoning paths and fine-grained control over retrieval. Experiments on LoCoMo and LongMemEval demonstrate that MAGMA consistently outperforms state-of-the-art agentic memory systems in long-horizon reasoning tasks. The open-source code is available at: <https://github.com/FredJiang0324/MAGMA>.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of tasks (Brown et al., 2020; Achiam et al., 2023; Wei et al., 2022), yet they remain limited in their ability to maintain and reason over long-term context. These models process information within a finite attention window, and their internal representations do not persist across interactions, causing earlier details to be forgotten once they fall outside the active context (Brown et al., 2020; Beltagy et al., 2020a). Even within a single long sequence, attention effectiveness degrades with distance due to *attention dilution*, positional encoding

limitations, and token interference, leading to the well-known “lost-in-the-middle” and context-decay phenomena (Liu et al., 2024; Press et al., 2021a). Moreover, LLMs lack native mechanisms for stable and structured memory, resulting in inconsistent recall, degraded long-horizon reasoning, and limited support for tasks requiring persistent and organized memory (Khandelwal et al., 2018; Maharana et al., 2024).

To address these inherent limitations, Memory-Augmented Generation (MAG) systems have emerged as a promising direction for enabling LLMs to operate beyond the boundaries of their fixed context windows. MAG equips an agent with an external memory continuously recording interaction histories and allowing the agents to retrieve and reintegrate past experiences when generating new responses. By offloading long-term context to an explicit memory module, MAG systems provide a means for agents to accumulate knowledge over time, support multi-session coherence, and adapt to evolving conversational or task contexts. In this paradigm, memory is no longer implicit in internal activations but becomes a persistent, queryable resource that substantially enhances long-horizon reasoning, personalized behavior, and stable agent identity.

Despite their promise, current MAG systems exhibit structural and operational limitations that constrain their effectiveness in long-term reasoning (Li et al., 2025; Chhikara et al., 2025; Xu et al., 2025; Packer et al., 2023; Rasmussen et al., 2025; Wang and Chen, 2025; Kang et al., 2025a). Most existing approaches store past interactions in monolithic repositories or minimally structured memory buffers, relying primarily on semantic similarity, recency, or heuristic scoring to retrieve relevant content. For example, A-Mem (Xu et al., 2025) organizes past interactions into Zettelkasten-like memory units that are incrementally linked and refined, yet their retrieval pipelines rely primarily on se-

MAGMA：一种面向AI智能体的基于多图的智能体记忆架构

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摘要

记忆增强生成（MAG）通过外部记忆扩展大语言模型以支持长上下文推理，但现有方法主要依赖于对单一化记忆存储的语义相似性搜索，将时序、因果和实体信息混杂在一起。这种设计限制了可解释性以及查询意图与检索证据之间的对齐，导致推理准确性欠佳。在本文中，我们提出 MAGMA，一种多图智能体记忆架构，它在正交的语义图、时序图、因果图和实体图中表示每个记忆项。MAGMA将检索过程形式化为在这些关系视图上进行策略引导遍历，从而实现查询自适应选择和结构化上下文构建。通过将记忆表示与检索逻辑解耦，MAGMA提供了透明的推理路径和对检索的细粒度控制。在LoCoMo和LongMemEval上的实验表明，MAGMA在长视野推理任务中持续优于最先进的智能体记忆系统。开源代码位于：

<https://github.com/FredJiang0324/MAGMA>。

1 引言

大语言模型（LLMs）已在广泛的任務中展现出卓越的能力（Brown 等人, 2020; Achiam 等人, 2023; Wei 等人, 2022），然而它们在维持和推理长期上下文方面的能力仍然有限。这些模型在有限的注意力窗口内处理信息，其内部表征不会在交互间持续存在，导致一旦早期细节超出活跃上下文就会被遗忘（Brown 等人, 2020; Belt-agy 等人, 2020a）。即使在单个长序列中，注意力机制的有效性也会因距离而降低，这是由于注意力稀释、位置编码

限制以及令牌干扰所致，从而导致了众所周知的“迷失在中间”和上下文衰减现象（Liu 等人, 2024; Press 等人, 2021a）。此外，大语言模型缺乏稳定且结构化的记忆的原生机制，导致回忆不一致、退化的长视野推理，以及对需要持久且有序的记忆的任务支持有限（Khandelwal 等人, 2018; Maharana 等人, 2024）。

为了应对这些固有的限制，记忆增强生成（MAG）系统已成为一个前景广阔的方向，它使大语言模型能够突破其固定上下文窗口的边界。MAG为智能体配备了外部记忆，持续记录交互历史，并允许智能体在生成新响应时检索并重新整合过去的经验。通过将长期上下文卸载到显式的记忆模块，MAG系统为智能体提供了一种随时间积累知识、支持多会话连贯性以及适应演进的对话或任务上下文的手段。在这种范式中，记忆不再隐含于内部激活中，而是成为一种持久、可查询的资源，从而极大地增强了长视野推理、个性化行为和稳定的智能体身份。

尽管前景广阔，但当前的MAG系统在结构和操作上存在局限性，制约了其在长期推理中的有效性（Li等人, 2025年; Chhikara等人, 2025年; Xu等人, 2025年; Packer等人, 2023年; Rasmussen等人, 2025年; Wang和Chen, 2025年; Kang等人, 2025a）。大多数现有方法将过去的交互存储在单一的整体存储库或结构极简的记忆缓冲区中，主要依赖语义相似性、时效性或启发式评分来检索相关内容。例如，A-Mem（Xu等人, 2025年）将过去的交互组织成类似Zettelkasten的记忆单元，这些单元会逐步链接和精炼，但其检索管道主要依赖于语

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mantic embedding similarity, missing the relations such as temporal or causal relationships. Cognitive-inspired frameworks like Nemori (Nan et al., 2025) introduce principled episodic segmentation and representation alignment, enabling agents to detect event boundaries and construct higher-level semantic summaries. However, their memory structures are still narrative and undifferentiated, with no explicit modeling of distinct relational dimensions.

To address the structural limitations of existing MAG systems, we propose MAGMA, a multi-graph agentic memory architecture that explicitly models heterogeneous relational structure in an agent’s experience. MAGMA represents each memory item across four orthogonal relational graphs (i.e., semantic, temporal, causal, and entity), yielding a disentangled representation of how events, concepts, and participants are related.

Built on this unified multi-graph substrate, MAGMA introduces a hierarchical, intent-aware query mechanism that selects relevant relational views, traverses them independently, and fuses the resulting subgraphs into a compact, type-aligned context for generation. By decoupling memory representation from retrieval logic, MAGMA enables transparent reasoning paths, fine-grained control over memory selection, and improved alignment between query intent and retrieved evidence. This relational formulation provides a principled and extensible foundation for agentic memory, improving both long-term coherence and interpretability.

Our contributions are summarized as follows:

1. We propose **MAGMA**, a multi-graph agentic memory architecture that explicitly models semantic, temporal, causal, and entity relations essential for long-horizon reasoning.
2. We introduce an Adaptive Traversal Policy that routes retrieval based on query intent, enabling efficient pruning of irrelevant graph regions and achieving lower latency and reduced token usage.
3. We design a dual-stream memory evolution mechanism that decouples latency-sensitive event ingestion from asynchronous structural consolidation, preserving responsiveness while refining relational structure.
4. We demonstrate that MAGMA consistently outperforms state-of-the-art agentic memory systems on long-context benchmarks including LoCoMo and LongMemEval, while reducing retrieval latency and token consumption relative

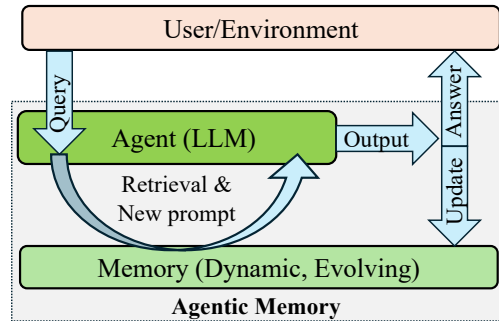


Figure 1: High-Level Architecture of Memory-Augmented Generation (MAG).

to prior systems. The code is open-sourced¹.

2 Background

Existing Large Language Models (LLMs) face fundamental challenges in handling long-term agentic interactions. These challenges stem from the inherent limitations of fixed-length contexts, which result in fragmented memory and an inability to maintain narrative coherence over time. The evolution of long-term consistency in LLMs is shifted from *Context-Window Extension* (Beltagy et al., 2020a; Press et al., 2021a; Kang et al., 2025c; Qian et al., 2025), *Retrieval-Augmented Generation (RAG)* (Lewis et al., 2020; Jiang et al., 2025; Wang et al., 2024; Jiang et al., 2024; Gutiérrez et al., 2025; Lin et al., 2025) to *Memory-Augmented Generation (MAG)*.

Retrieval-oriented approaches enrich the model with an external, dynamic memory library, giving rise to the paradigm of Memory-Augmented Generation (MAG) (Zhong et al., 2024; Park et al., 2023; Huang et al., 2024). Formally, unlike static RAG, MAG maintains a time-variant memory $\mathcal{M}_t$ that evolves via a feedback loop:

$$o_t = \text{LLM}(q_t, \text{Retrieve}(q_t, \mathcal{M}_t)) \quad (1)$$

$$\mathcal{M}_{t+1} = \text{Update}(\mathcal{M}_t, q_t, o_t) \quad (2)$$

As shown in Figure 1, this feedback loop enables the memory module to evolve over time: the user query is combined with retrieved information to form an augmented prompt, and the model’s output is subsequently written back to refine $\mathcal{M}_t$.

Some prior schemes focused on structuring the intermediate states or relationships of memory to enable better reasoning. Think-in-Memory (TiM) (Liu et al., 2023) stores evolving chains-of-thought

¹<https://github.com/FredJiang0324/MAGMA>

义嵌入相似性，忽略了诸如时序或因果关系等关系。像Nemori (Nan等人, 2025年) 这类受认知启发的框架引入了原则性的事件分割和表征对齐，使智能体能够检测事件边界并构建更高层次的语义摘要。然而，它们的记忆结构仍然是叙事性的且未加区分，没有对不同关系维度进行显式建模。

为解决现有MAG系统的结构局限性，我们提出了MAGMA，一种多图智能体记忆架构，它显式地对智能体经验中的异构关系结构进行建模。MAGMA将每个记忆项在四个正交关系图（即语义、时序、因果和实体图）中进行表征，从而得到关于事件、概念和参与者如何关联的解耦表征。

基于此统一多图基底，MAGMA引入了一种分层意图感知查询机制，该机制选择相关的关系视图，独立遍历它们，并将生成的子图融合成一个紧凑的、类型对齐上下文以供生成。通过将记忆表征与检索逻辑解耦，MAGMA实现了透明的推理路径、对记忆选择的细粒度控制，以及查询意图与检索证据之间更好的对齐。这种关系表述为智能体记忆提供了一个原则性且可扩展的基础，同时提升了长期连贯性和可解释性。

我们的贡献总结如下：

1. 我们提出了 **MAGMA**，一种多图智能体记忆架构，它显式地对语义、时序、因果和实体关系进行建模，这些关系对于长视野推理至关重要。
2. 我们引入了一种自适应遍历策略，该策略根据查询意图来路由检索，从而能够高效剪枝不相关的图区域，并实现更低的延迟和更少的令牌使用。
3. 我们设计了一种双流记忆演化机制，它将延迟敏感事件摄取与异步结构整合解耦，在优化关系结构的同时保持了响应性。
4. 我们证明，在包括LoCoMo和LongMemEval在内的长上下文基准测试中，MAGMA始终优于最先进的智能体记忆系统，同时相对于

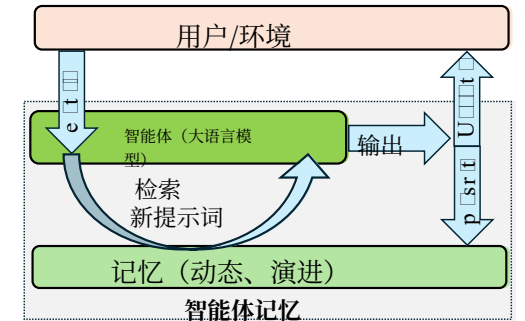


图 1: 记忆增强生成 (MAG) 的高层架构。

先前的系统降低了检索延迟和令牌消耗。代码已开源¹。

2 背景

现有的大语言模型在处理长期智能体交互时面临根本性挑战。这些挑战源于固定长度上下文的固有局限，导致碎片化记忆以及无法随时间推移保持叙事连贯性。大语言模型中长期一致性的演进，已从上下文窗口扩展 (Beltagy等人, 2020a; Press等人, 2021a; Kang等人, 2025c; Qian等人, 2025)、检索增强生成 (Lewis等人, 2020; Jiang等人, 2025; Wang等人, 2024; Jiang等人, 2024; Gutiérrez等人, 2025; Lin等人, 2025) 转向记忆增强生成。

面向检索的方法通过一个外部动态记忆库来丰富模型，催生了记忆增强生成范式 (Zhong等人, 2024; Park等人, 2023; Huang等人, 2024)。形式上，与静态 RAG 不同，MAG 维护一个时变记忆 $\mathcal{M}_t$ ，该记忆通过反馈循环不断演进：

$$o_t = \text{LLM}(q_t, \text{Retrieve}(q_t, \mathcal{M}_t)) \quad (1)$$

$$\mathcal{M}_{t+1} = \text{Update}(\mathcal{M}_t, q_t, o_t) \quad (2)$$

如图1所示，此反馈循环使得记忆模块能够随时间演进：用户查询与检索信息结合形成增强提示词，随后模型的输出会被写回以优化 $\mathcal{M}_t$ 。

一些先前方案侧重于构建记忆的中间状态或关系，以实现更好的推理。内存思考 (TiM) (Liu et al., 2023) 存储不断演化的思维链

¹<https://github.com/FredJiang0324/MAGMA>

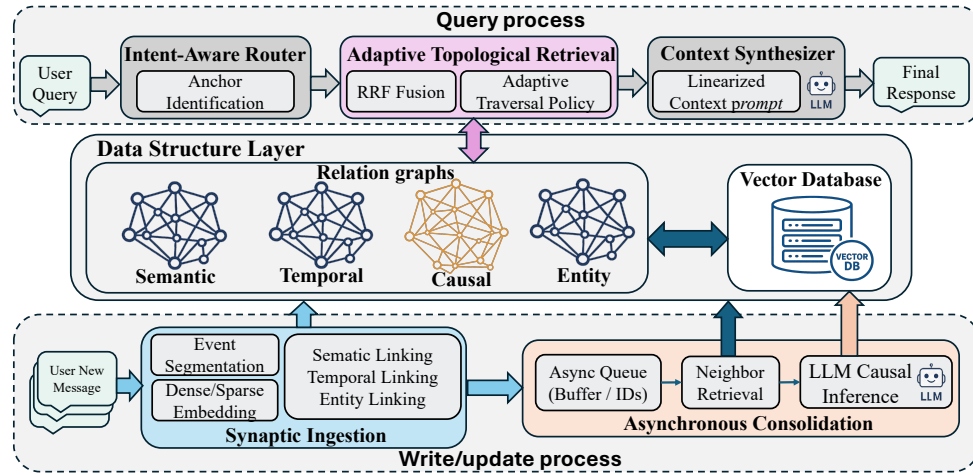


Figure 2: Architectural Overview of MAGMA. The system is composed of three layers: (1) A Query Process that routes and synthesizes context; (2) A Data Structure Layer organizing memory into Relation Graphs and a Vector Database; and (3) A Write/Update Process utilizing a dual-stream mechanism for fast ingestion and asynchronous consolidation.

to maintain consistency. A-MEM (Xu et al., 2025) draws inspiration from the Zettelkasten method, organizing knowledge into an interconnected note network. More recently, graph-based approaches like GraphRAG (Edge et al., 2024a) and Zep (Rasmussen et al., 2025) structure memory into knowledge graphs to capture cross-document dependencies. We provide a detailed discussion of related work in Appendix A.

However, prior work typically organizes memory around associative proximity (e.g., semantic similarity) rather than mechanistic dependency (Kiciman et al., 2023). As a result, such methods can retrieve *what* occurred but struggle to reason about *why*, since they lack explicit representations of causal structure, leading to reduced accuracy in complex reasoning tasks (Jin et al., 2023; Zhang et al., 2025).

3 MAGMA Design

In this section, we introduce the proposed Multi-Graph based Agentic Memory (MAGMA) design and its components in detail.

3.1 Architectural Overview

MAGMA architecture is organized into the following three logical layers, orchestrating the interaction between control logic and the memory substrate as illustrated in Figure 2.

Query Process: The inference engine responsible for retrieving and synthesizing information. It comprises the *Intent-Aware Router* for dispatching

tasks, the *Adaptive Topological Retrieval* module for executing graph traversals, and the *Context Synthesizer* for generating the final narrative response.

Data Structure (G): The unified storage substrate that fuses disparate modalities. As shown in the center of Figure 2, it maintains a *Vector Database* for semantic search alongside four distinct *Relation Graphs* (i.e., Semantic, Temporal, Causal and Entity). This layer provides the topological foundation for cross-view reasoning.

Write/Update Process: A dual-stream pipeline manages memory evolution. It decouples latency-sensitive operations via *Synaptic Ingestion* (Fast Path) from compute-intensive reasoning via *Asynchronous Consolidation* (Slow Path), ensuring the system remains responsive while continuously deepening its memory structure.

Functionally, the Query Layer interacts with the Data Structure Layer to execute the synchronous Query Process (Section 3.3), while the Write/Update Layer manages the continuous Memory Evolution (Section 3.4).

3.2 Data Structure Layer

As the core component of Memory-Augmented Generation (MAG), the data structure layer is responsible for storing, organizing, and evolving past information to support future retrieval and updates. In MAGMA, we formalize this layer as a time-variant directed multigraph $\mathcal{G}_t = (\mathcal{N}_t, \mathcal{E}_t)$, where nodes represent events and edges encode heterogeneous relational structures. This unified manifold

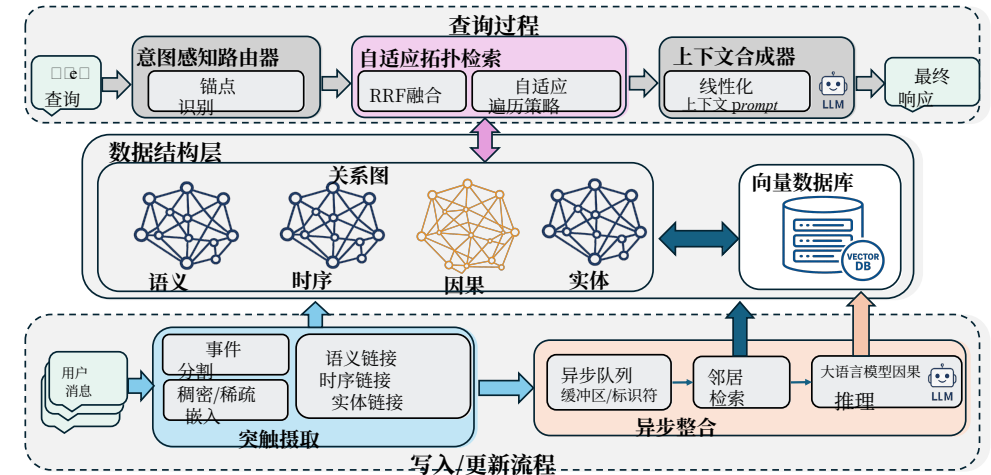


图 2: MAGMA 架构概览。该系统由三层组成：(1) 一个用于路由和合成上下文的查询处理器；(2) 一个将记忆组织为关系图和向量数据库的数据结构层；(3) 一个利用双流机制进行快速摄取和异步整合的写入/更新流程。

以保持一致性。A-Mem (Xu et al., 2025) 从卡片盒笔记法中汲取灵感，将知识组织成互连笔记网络。最近，基于图的方法如 GraphRAG (Edge et al., 2024a) 和 Zep (Rasmussen et al., 2025) 将记忆构建为知识图，以捕捉跨文档依赖关系。我们在附录 A 中提供了相关工作的详细讨论。

然而，先前的工作通常围绕关联邻近性（例如，语义相似性）而非机制依赖性来组织记忆 (Kiciman 等人, 2023)。因此，这类方法能够检索发生了什么，但难以推理原因，因为它们缺乏对因果结构的显式表示，导致在复杂推理任务中准确性降低 (Jin 等人, 2023; Zhang 等人, 2025)。

3 MAGMA 设计

在本节中，我们将详细介绍所提出的基于多图的智能体记忆 (MAGMA) 设计及其组件。

3.1 架构概述

MAGMA 架构组织为以下三个逻辑层，协调控制逻辑与记忆基底之间的交互，如图 2 所示。

查询过程：负责检索与综合信息的推理引擎。它包含用于分派任务的意图感知路由器，

用于执行图遍历的自适应拓扑检索模块，以及用于生成最终叙述性响应的上下文综合器。**数据结构 (G)：**融合异构模态的统一存储基底。如图 2 中心所示，它维护一个用于语义搜索的向量数据库，以及四个不同的关系图（即语义、时序、因果和实体图）。该层为跨视图推理提供了拓扑基础。

写入/更新流程：一个双流管道管理记忆演化。它通过突触摄取（快速路径）将延迟敏感的操作与通过异步巩固（慢速路径）进行的计算密集型推理解耦，确保系统保持响应性的同时，持续深化其记忆结构。

在功能上，查询层与数据结构层交互以执行同步查询过程（第 3.3 节），而写入/更新层则管理持续的记忆演化（第 3.4 节）。

3.2 数据结构层

作为记忆增强生成 (MAG) 的核心组件，数据结构层负责存储、组织和演化过去的信息，以支持未来的检索和更新。在 MAGMA 中，我们将此层形式化为一个时变有向多重图 $\mathcal{G}_t = (\mathcal{N}_t, \mathcal{E}_t)$ ，其中节点表示事件，边编码异构的关系结构。这个统一的流形

enables structured reasoning across multiple logical dimensions(i.e., semantic, temporal, causal, and entity) while preserving their orthogonality.

Unified node representation: The node set $\mathcal{N}$ is hierarchically organized to represent experience at multiple granularities, ranging from fine-grained atomic events to higher-level episodic groupings. Each Event-Node $n_i \in \mathcal{N}_{\text{event}}$ is defined as:

$$n_i = \langle c_i, \tau_i, \mathbf{v}_i, \mathcal{A}_i \rangle \quad (3)$$

where c_i denotes the event content (e.g., observations, actions, or state changes), τ_i is a discrete timestamp anchoring the event in time, and $\mathbf{v}_i \in \mathbb{R}^d$ is a dense representation indexed in the vector database (Johnson et al., 2019). The attribute set $\mathcal{A}_i$ captures structured metadata such as entity references, temporal cues, or contextual descriptors, enabling hybrid retrieval that integrates semantic similarity with symbolic and structural constraints.

Relation graphs (edge space): The edge set $\mathcal{E}$ is partitioned into four semantic subspaces, corresponding to the relation graphs:

- **Temporal Graph ($\mathcal{E}_{temp}$):** Defined as strictly ordered pairs (n_i, n_j) where $\tau_i < \tau_j$. This immutable chain provides the ground truth for chronological reasoning.
- **Causal Graph ($\mathcal{E}_{causal}$):** Directed edges representing logical entailment. An edge $e_{ij} \in \mathcal{E}_{causal}$ exists if $S(n_j|n_i, q) > \delta$, explicitly inferred by the consolidation module to support "Why" queries.
- **Semantic Graph ($\mathcal{E}_{sem}$):** Undirected edges connecting conceptually similar events, formally defined by $\cos(\mathbf{v}_i, \mathbf{v}_j) > \theta_{sim}$.
- **Entity Graph ($\mathcal{E}_{ent}$):** Edges connecting events to abstract entity nodes, solving the object permanence problem across disjoint timeline segments.

3.3 Query Process: Adaptive Hierarchical Retrieval

As illustrated in Figure 3, retrieval in MAGMA is formulated as a policy-guided graph traversal rather than a static lookup operation. The query process is orchestrated by a Router $\mathcal{R}$, which decomposes the user query into structured control signals and executes a multi-stage retrieval pipeline (Algorithm 1) that dynamically selects, traverses, and fuses relevant relational views. Four main stages in the query process is introduced below:

Stage 1 - Query Analysis & Decomposition: The process begins by decomposing the raw user query q into structured control signals, including semantic, lexical, and temporal cues. MAGMA then extracts three complementary representations to guide the retrieval process:

- **Intent Classification (T_q):** A lightweight classifier maps q to a specific intent type $T_q \in \{\text{WHY, WHEN, ENTITY}\}$. This acts as the "steering wheel," determining which graph edges will later be prioritized (e.g., "Why" queries trigger a bias for Causal edges).
- **Temporal Parsing ($[\tau_s, \tau_e]$):** A temporal tagger resolves relative expressions (e.g., "last Friday") into absolute timestamps, defining a hard time window for filtering.
- **Representation Extraction:** The system simultaneously generates a dense embedding $\vec{q}$ for semantic search and extracts sparse keywords q_{key} for exact lexical matching.

Stage 2 - Multi-Signal Anchor Identification: Before initiating graph traversal, the system first identifies a set of anchor nodes that serve as entry points into the memory graph. To ensure robustness across query modalities, we fuse signals from dense semantic retrieval, lexical keyword matching, and temporal filtering using Reciprocal Rank Fusion (RRF) (Cormack et al., 2009):

$$S_{anchor} = \text{Top}_K \left(\sum_{m \in \{vec, key, time\}} \frac{1}{k + r_m(n)} \right) \quad (4)$$

This ensures robust starting points regardless of query modality.

Stage 3 - Adaptive Traversal Policy: Starting from the anchor set S_{anchor} , the system expands the context using a Heuristic Beam Search. Unlike rigid rule-based traversals, MAGMA calculates a dynamic transition score $S(n_j|n_i, q)$ for moving from node n_i to neighbor n_j via edge e_{ij} . This score fuses structural alignment with semantic relevance:

$$S(n_j|n_i, q) = \exp \left(\underbrace{\lambda_1 \cdot \phi(\text{type}(e_{ij}), T_q)}_{\text{Structural Alignment}} + \underbrace{\lambda_2 \cdot \text{sim}(\vec{n}_j, \vec{q})}_{\text{Semantic Affinity}} \right) \quad (5)$$

支持跨多个逻辑维度（即语义、时序、因果和实体）的结构化推理，同时保持它们的正交性。

统一节点表示: 节点集 $\mathcal{N}$ 采用分层组织，以表示从细粒度原子事件到更高级别情景分组的多粒度经验。每个事件节点 $n_i \in \mathcal{N}_{\text{event}}$ 定义为:

$$n_i = \langle c_i, \tau_i, \mathbf{v}_i, \mathcal{A}_i \rangle \quad (3)$$

其中 c_i 表示事件内容（例如，观察、动作或状态变化）， τ_i 是一个将事件锚定在时间中的离散时间戳，而 $\mathbf{v}_i \in \mathbb{R}^d$ 是索引在向量数据库中的稠密表示 (Johnson 等人, 2019)。属性集 $\mathcal{A}_i$ 捕获结构化元数据，如实体引用、时序线索或上下文描述符，从而实现融合语义相似性与符号及结构约束的混合检索。

关系图（边空间）: 边集 $\mathcal{E}$ 被划分为四个语义子空间，分别对应以下关系图:

- **时序图 ($\mathcal{E}_{temp}$):** 定义为严格有序对 (n_i, n_j) ，其中 $\tau_i < \tau_j$ 。这个不可变的链为时序推理提供了基本事实依据。
- **因果图 ($\mathcal{E}_{causal}$):** 有向边表示逻辑蕴含。若 $S(n_j|n_i, q) > \delta$ ，则存在边 $e_{ij} \in \mathcal{E}_{causal}$ ，由整合模块显式推断以支持“Why”查询。
- **语义图 ($\mathcal{E}_{sem}$):** 无向边连接概念相似事件，形式化定义为 $\cos(\mathbf{v}_i, \mathbf{v}_j) > \theta_{sim}$ 。
- **实体图 ($\mathcal{E}_{ent}$):** 边连接事件与抽象实体节点，解决跨不相交时间线片段的物体恒存问题。

3.3 查询过程：自适应分层检索

如图3所示，MAGMA中的检索被表述为一种策略引导的图遍历，而非静态查找操作。查询过程由Router $\mathcal{R}$ 编排，它将用户查询分解为结构化控制信号，并执行一个多阶段检索流水线（算法1），动态选择、遍历并融合相关的关系视图。查询过程的四个主要阶段介绍如下:

阶段 1 - 查询分析与分解: 该过程始于将原始用户查询 q 分解为结构化控制信号，包括语义、词汇和时序线索。随后，MAGMA提取三种互补的表征来引导检索过程:

- **意图分类 (T_q):** 轻量级分类器将 q 映射到特定意图类型 $T_q \in \{\text{WHY, WHEN, ENTITY}\}$ 。这充当“方向盘”，决定后续将优先考虑哪些图边（例如，“Why”查询会触发对因果边的偏好）。
- **时序解析 ($[\tau_s, \tau_e]$):** 时序标注器将相对表达（例如“上周五”）解析为绝对时间戳，从而定义一个用于过滤的硬时间窗口。
- **表征提取:** 系统同时生成一个用于语义搜索的稠密嵌入 $\vec{q}$ ，并提取用于精确词汇匹配的稀疏关键词 q_{key} 。

阶段 2 - 多信号锚点识别: 在启动图遍历之前，系统首先识别一组锚点节点，作为进入记忆图的入口点。为确保跨查询模态的鲁棒性，我们使用互逆排序融合 (RRF) (科马克等人, 2009) 融合来自稠密语义检索、词汇关键词匹配和时间过滤的信号:

$$S_{anchor} = \text{Top}_K \left(\sum_{m \in \{vec, key, time\}} \frac{1}{k + r_m(n)} \right) \quad (4)$$

这确保了无论查询模态如何，都能获得鲁棒的起始点。

阶段 3 - 自适应遍历策略: 从锚点集 S_{anchor} 出发，系统使用启发式束搜索来扩展上下文。与僵化的基于规则的遍历不同，MAGMA 会计算一个动态的转移分数 $S(n_j|n_i, q)$ ，用于通过边 e_{ij} 从节点 n_i 移动到邻居 n_j 。该分数融合了结构对齐与语义相关性:

$$S(n_j|n_i, q) = \exp \left(\underbrace{\lambda_1 \cdot \phi(\text{type}(e_{ij}), T_q)}_{\text{Structural Alignment}} + \underbrace{\lambda_2 \cdot \text{sim}(\vec{n}_j, \vec{q})}_{\text{Semantic Affinity}} \right) \quad (5)$$

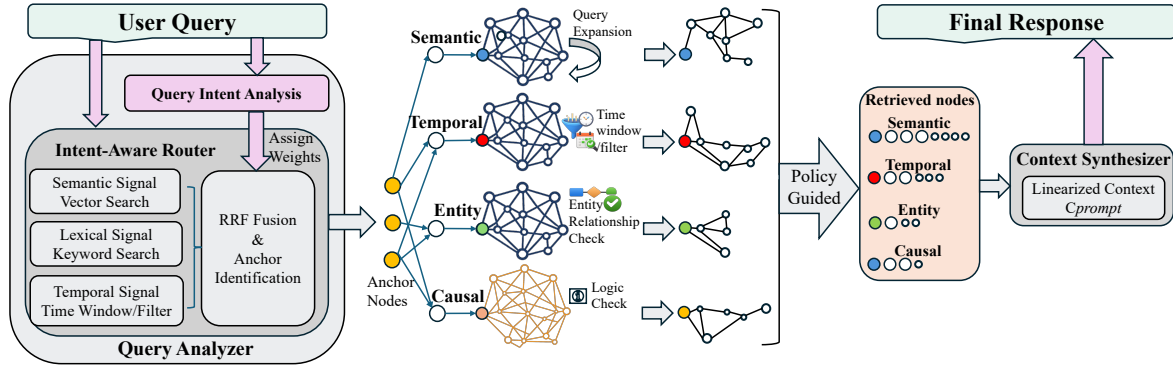


Figure 3: Query process with adaptive hybrid retrieval pipeline. (1) Query Analysis detects intent and fuses signals to find Anchors. (2) Adaptive Traversal navigates specific graph views (Causal, Temporal) based on the policy weights.

Here, $\text{sim}(\cdot)$ denotes the cosine similarity between the neighbor's embedding and the query embedding. The structural alignment function ϕ dynamically rewards edge types based on the detected query intent T_q :

$$\phi(r, T_q) = \mathbf{w}_{T_q}^\top \cdot \mathbf{1}_r \quad (6)$$

where $\mathbf{w}_{T_q}$ is an adaptive weight vector specific to intent T_q (e.g., assigning high weights to CAUSAL edges for "Why" queries), and $\mathbf{1}_r$ is the one-hot encoding of the edge relation.

At each step, the algorithm retains the top- k nodes with the highest cumulative scores. This ensures the traversal is guided by a dual signal: strictly following the logical structure (via ϕ) while maintaining contextual focus (via sim).

Stage 4: Narrative Synthesis via Graph Linearization: The final phase transforms the retrieved subgraph $\mathcal{G}_{sub}$ into a coherent narrative context. MAGMA employs a structure-aware linearization protocol that preserves the relational dependencies encoded in the graph with the following three phases.

1. Topological Ordering: Raw nodes are reorganized to reflect the logic of the query. For temporal queries ($T_q = \text{WHEN}$), nodes are sorted by timestamp τ_i . For causal queries ($T_q = \text{WHY}$), we apply a topological sort on the causal edges $\mathcal{E}_{causal}$ to ensure causes precede effects in the prompt context.

2. Context Scaffolding with Provenance: To mitigate hallucination, each node is serialized into a structured block containing its timestamp, content, and explicit reference ID. We define the linearized

Algorithm 1 Adaptive Hybrid Retrieval (Heuristic Beam Search)

Input: Query q , Graph G , VectorDB V , Intent T_q
Output: Narrative Context C_{out}

```

1: // Phase 1: Initialization
2:  $S_{anchor} \leftarrow \text{RRF}(V.\text{SEARCH}(\vec{q}), K.\text{SEARCH}(q_{key}))$  // Hybrid Retrieval
3:  $CurrentFrontier, Visited \leftarrow S_{anchor}$ 
4:  $\mathbf{w}_{T_q} \leftarrow \text{GETATTENTIONWEIGHTS}(T_q)$ 
5: for  $d \leftarrow 1$  to  $MaxDepth$  do
6:    $Candidates \leftarrow \text{PRIORITYQUEUE}()$ 
7:   for  $u \in CurrentFrontier$  do
8:     for  $v \in G.\text{NEIGHBORS}(u)$  do
9:       if  $v \notin Visited$  then
10:        // Calculate transition score via Eq. 5
11:         $s_{uv} \leftarrow \exp(\lambda_1(\mathbf{w}_{T_q}^\top \cdot \mathbf{1}_{e_{uv}}) + \lambda_2 \text{sim}(\vec{v}, \vec{q}))$ 
12:         $score_v \leftarrow score_u \cdot \gamma + s_{uv}$  // Apply Decay  $\gamma$ 
13:         $Candidates.\text{PUSH}(v, score_v)$ 
14:       end if
15:     end for
16:   end for
17:    $CurrentFrontier \leftarrow Candidates.\text{TOPK}(BeamWidth)$ 
18:    $Visited.\text{ADDALL}(CurrentFrontier)$ 
19:   if  $Visited.\text{SIZE}() \geq Budget$  then break
20:   end if
21: end for
22:  $C_{sorted} \leftarrow \text{TOPOLOGICALSORT}(Visited, T_q)$ 
23: return  $\text{SERIALIZE}(C_{sorted})$ 

```

context C_{prompt} as:

$$C_{prompt} = \bigoplus_{n_i \in \text{Sort}(\mathcal{G}_{sub})} [\langle t: \tau_i \rangle \ n_i.\text{content} \ \langle \text{ref}: n_i.\text{id} \rangle] \quad (7)$$

where $\bigoplus$ denotes string concatenation.

3. Salience-Based Token Budgeting: Given a fixed LLM context window, we cannot include all retrieved nodes. We utilize the relevance scores $S(n_j|n_i, q)$ computed in Eq. (5) to enforce a dynamic budget. Low-probability nodes are summarized into brevity codes (e.g., "...3 intermediate

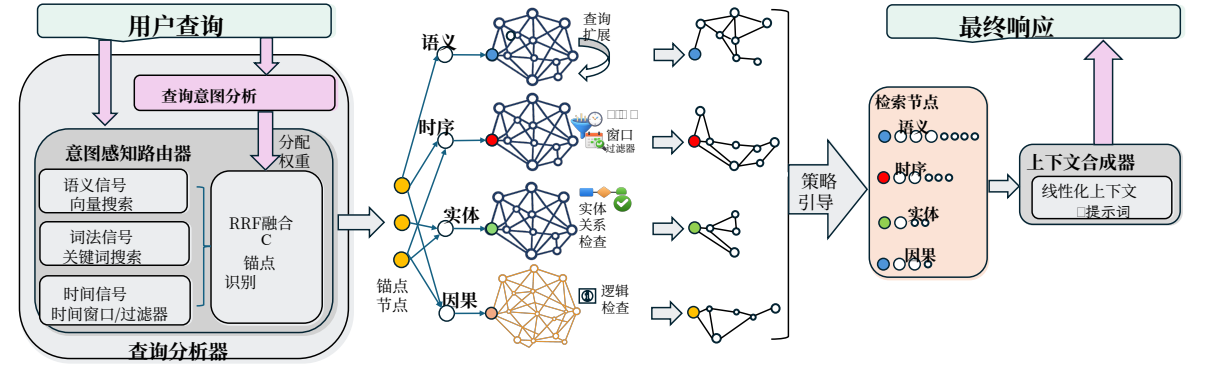


图 3: 采用自适应混合检索管道的查询过程。(1) 查询分析检测意图并融合信号以找到锚点。(2) 自适应遍历根据策略权重在特定图视图（因果、时序）中进行导航。

此处, $\text{sim}(\cdot)$ 表示邻居嵌入与查询嵌入之间的余弦相似度。结构对齐函数 ϕ 会根据检测到的查询意图 T_q 动态地为边类型赋予奖励:

$$\phi(r, T_q) = \mathbf{w}_{T_q}^\top \cdot \mathbf{1}_r \quad (6)$$

其中 $\mathbf{w}_{T_q}$ 是一个针对特定意图 T_q 的自适应权重向量（例如，为“为什么”查询分配高权重给因果边），而 $\mathbf{1}_r$ 是边关系的独热编码。

在每一步，算法保留累积分数最高的前 k 个节点。这确保了遍历由双重信号引导：严格遵循逻辑结构（通过 ϕ ），同时保持上下文焦点（通过相似度）。

阶段 4：通过图线性化进行叙事合成：最终阶段将检索到的子图 $\mathcal{G}_{sub}$ 转化为连贯的叙事上下文。MAGMA 采用一种结构感知的线性化协议，该协议通过以下三个阶段保留图中编码的关系依赖。

1. 拓扑排序：原始节点被重新组织以反映查询的逻辑。对于时间查询（ $T_q = \text{WHEN}$ ），节点按时间戳 τ_i 排序。对于因果查询（ $T_q = \text{WHY}$ ），我们对因果边 $\mathcal{E}_{causal}$ 应用拓扑排序，以确保在提示词上下文中原因先于效果出现。

2. 带有来源的上下文支架：为减轻幻觉，每个节点都被序列化为一个包含其时间戳、内容和显式参考ID的结构化块。我们定义线性化

算法1 自适应混合检索（启发式束搜索）

输入： 查询 q , 图 G , 向量数据库 V , 意图 T_q
输出： 叙事上下文 C_{out}

```

1: // 第一阶段：初始化
2:  $S_{anchor} \leftarrow \text{RRF}(V.\text{搜索}(\vec{q}) \ K.\text{搜索}(q_{key}))$  // 混合检索
3:  $CurrentFrontier, Visited \leftarrow S_{anchor}$ 
4: 获取  $T_q \leftarrow$  图注意力权重( $T_q$ )
5: 对于  $d \leftarrow 1$  到  $MaxDepth$  执行
6:    $Candidates \leftarrow$  优先级队列()
7:   for  $u \in CurrentFrontier$  do
8:     对于  $v \in G.\text{邻居}(u)$  执行
9:       如果  $v \notin Visited$  则
10:        // 通过公式5计算转移分数
11:         $s_{uv} \leftarrow \exp(\lambda_1(\mathbf{w}_{T_q}^\top \cdot \mathbf{1}_{e_{uv}}) + \lambda_2 \text{相似度}(\vec{v}, \vec{q}))$ 
12:         $score_v \leftarrow \text{分数}_u \cdot \gamma + s_{uv}$  // 应用衰减  $\gamma$ 
13:         $Candidates.\text{推送}(v, score_v)$ 
14:       结束如果
15:     结束循环
16:   结束循环
17:    $CurrentFrontier \leftarrow Candidates.\text{TOPK}(BeamWidth)$ 
18:    $Visited.\text{动态链接}(CurrentFrontier)$ 
19:   如果  $Visited.\text{SIZE}() \geq Budget$  则跳出循环
20:   结束条件判断
21: 结束循环
22:  $C_{sorted} \leftarrow$  拓扑排序( $Visited, T_q$ )
23: 返回序列化( $C_{sorted}$ )

```

上下文 C_{prompt} 为:

$$C_{prompt} = \bigoplus_{n_i \in \text{Sort}(\mathcal{G}_{sub})} [\langle t: \tau_i \rangle \ n_i.\text{content} \ \langle \text{ref}: n_i.\text{id} \rangle] \quad (7)$$

其中 $\bigoplus$ 表示字符串拼接。

3. 基于显著性的令牌预算：给定一个固定的大语言模型上下文窗口，我们无法包含所有检索节点。我们利用公式(5)中计算出的相关度分数 $S(n_j|n_i, q)$ 来实施动态预算。低概率节点被总结为简洁代码（例如，“...3个中间

Algorithm 2 Fast Path: Synaptic Ingestion

Input: User Interaction I , Current Graph $\mathcal{G}_t$
Output: Updated Graph $\mathcal{G}_{t+1}$

```

1:  $n_t \leftarrow \text{SEGMENTEVENT}(I)$ 
2:  $n_{prev} \leftarrow \text{GETLASTNODE}(\mathcal{G}_t)$ 
3:                                     // Update Temporal Backbone
4:  $\mathcal{G}.\text{ADDEDGE}(n_{prev}, n_t, \text{type} = \text{TEMP})$ 
5:                                     // Indexing
6:  $\mathbf{v}_t \leftarrow \text{ENCODER}(n_t.c)$ 
7:  $VDB.\text{ADD}(\mathbf{v}_t, n_t.id)$ 
8:  $Queue.\text{ENQUEUE}(n_t.id)$            // Trigger Slow Path
9: return  $n_t$ 

```

events..."), while high-salience nodes retain full semantic detail.

This structured scaffold forces the LLM to act as an interpreter of evidence rather than a creative writer, significantly reducing grounding errors.

3.4 Memory Evolution (Write and Update)

Long-term reasoning requires not only effective retrieval, but also a memory substrate that can adapt and reorganize as experience accumulates. MAGMA addresses this requirement through a structured memory evolution scheme that incrementally refines its multi-relational graph over time. Specifically, the transition from $\mathcal{G}_t$ to $\mathcal{G}_{t+1}$ is governed by a dual-stream process that decouples latency-sensitive ingestion from compute-intensive consolidation (Kumaran et al., 2016), balancing short-term responsiveness with long-term reasoning fidelity.

Fast path (synaptic ingestion): The Fast Path operates on the critical path of interaction, constrained by strict latency requirements. It performs non-blocking operations: event segmentation, vector indexing, and updating the immutable temporal backbone ($n_{t-1} \rightarrow n_t$). As detailed in Algorithm 2, no blocking LLM reasoning occurs here, ensuring the agent remains responsive regardless of memory size.

Slow path (structural consolidation): Asynchronously, the slow path performs Memory Consolidation (Algorithm 3). It functions as a background worker that dequeues events and densifies the graph structure. By analyzing the local neighborhood $\mathcal{N}(n_t)$ of recent events, the system employs an LLM Φ to infer latent connections:

$$\mathcal{E}_{new} = \Phi_{reason}(\mathcal{N}(n_t), \mathcal{H}_{history}) \quad (8)$$

This process constructs high-value $\mathcal{E}_{causal}$ and $\mathcal{E}_{ent}$ links, effectively trading off compute time for relational depth.

Algorithm 3 Slow Path: Structural Consolidation

```

1: Worker Process:
2: loop
3:    $id \leftarrow Queue.\text{DEQUEUE}()$ 
4:   if  $id$  is null then continue
5:   end if
6:    $n_t \leftarrow \mathcal{G}.\text{GETNODE}(id)$ 
7:    $\mathcal{N}_{local} \leftarrow \mathcal{G}.\text{GETNEIGHBORHOOD}(n_t, \text{hops} = 2)$ 
8:                                       // Infer latent Causal and Entity structures
9:    $Prompt \leftarrow \text{FORMAT}(\mathcal{N}_{local})$ 
10:   $\mathcal{E}_{new} \leftarrow \Phi_{LLM}(Prompt)$ 
11:   $\mathcal{G}.\text{ADDEDGES}(\mathcal{E}_{new})$ 
12: end loop

```

3.5 Implementation

We implement MAGMA as a modular three-layer architecture designed for extensibility, scalability, and deployment flexibility. The storage layer abstracts over heterogeneous physical backends, providing unified interfaces for managing the typed memory graph, dense vector indices, and sparse keyword indices. This abstraction cleanly separates the logical memory model from its physical realization, enabling seamless substitution of storage backends (e.g., in-memory data structures versus production-grade graph or vector databases) with minimal engineering effort.

The retrieval layer coordinates the core algorithmic components, including memory construction, multi-stage ranking, and policy-guided graph traversal. It is supported by specialized utility modules for episodic segmentation and temporal normalization, which provide structured signals to downstream retrieval and traversal policies. The application layer manages the interaction loop, evaluation harnesses, and prompt construction, serving as the interface between the agent and the underlying memory system.

4 Experiments

We conduct comprehensive experiments to evaluate both the reasoning effectiveness and systems properties of the proposed MAGMA architecture over state-of-the-art baselines.

4.1 Experimental Setup

Datasets. We evaluate long-term conversational capability using two widely adopted benchmarks: **(1) LoCoMo (Maharana et al., 2024):** which contains ultra-long conversations (average length of 9K tokens) designed to assess long-range temporal and causal retrieval. **(2) LongMemEval (Wu et al., 2024):** a large-scale stress-test benchmark with

算法 2 快速路径：突触摄取

输入: 用户交互 I , 当前图 $\mathcal{G}_t$
输出: 更新图 $\mathcal{G}_{t+1}$

```

1:  $n_t \leftarrow \text{分段事件}(I)$ 
2:  $n_{prev} \leftarrow \text{获取最后节点}(\mathcal{G}_t)$ 
3:                                     // 更新时间骨干
4:  $\mathcal{G}.\text{添加边}(n_{prev}, n_t, \text{类型} = \text{TEMP})$ 
5:                                     // 索引
6:  $\mathbf{v}_t \leftarrow \text{E编码器}(n_t.c)$ 
7:  $VDB.\text{添加}(\mathbf{v}_t, n_t.id)$ 
8:  $Queue.\text{E入队列}(n_t.id)$            // 触发慢速路径
9: 返回  $n_t$ 

```

事件...”），而高显著性节点则保留完整的语义细节。

这种结构化框架强制大语言模型充当证据的解释者，而非创意写手，从而显著减少了事实依据错误。

3.4 记忆演化（写入与更新）

长期推理不仅需要有效的检索，还需要一个能够随着经验积累而适应和重组的记忆基底。MAGMA通过一种结构化的记忆演化方案来满足这一需求，该方案会随时间推移逐步完善其多关系图。具体而言，从 $\mathcal{G}_t$ 到 $\mathcal{G}_{t+1}$ 的转换由一个双流过程控制，该过程将延迟敏感的摄取与计算密集的巩固解耦（Kumaran 等人, 2016），从而在短期响应能力和长期推理保真度之间取得平衡。

快速路径（突触摄取）：快速路径在交互的关键路径上运行，受严格的延迟要求约束。它执行非阻塞操作：事件分割、向量索引以及更新不可变的时间骨干（ $n_{t-1} \rightarrow n_t$ ）。如算法2所述，此处不发生阻塞性的大语言模型推理，确保智能体无论记忆规模大小都能保持响应性。

慢速路径（结构巩固）：异步地，慢速路径执行记忆巩固（算法3）。它作为后台工作进程，将事件出队并致密化图结构。通过分析局部邻域 $\mathcal{N}(n_t)$ 的近期事件，系统利用大语言模型 Φ 来推断潜在连接：

$$\mathcal{E}_{new} = \Phi_{reason}(\mathcal{N}(n_t), \mathcal{H}_{history}) \quad (8)$$

此过程构建高价值的 $\mathcal{E}_{causal}$ 和 $\mathcal{E}_{ent}$ 链接，有效地以计算时间换取关系深度。

算法3 慢速路径：结构巩固

```

1: 工作进程:
2: 循环
3:    $id \leftarrow Queue.\text{出队}()$ 
4:   如果  $id$  为空 则继续
5:   结束如果
6:    $n_t \leftarrow \mathcal{G}.\text{图获取节点操作}(id)$ 
7:    $\mathcal{N}_{local} \leftarrow \mathcal{G}.\text{获取邻域}(n_t, \text{跳数} = 2)$ 
8:                                       // 推断潜在因果与实体结构
9:    $Prompt \leftarrow \text{格式化}(\mathcal{N}_{local})$ 
10:   $\mathcal{E}_{new} \leftarrow \Phi_{LLM}(Prompt)$ 
11:   $\mathcal{G}.\text{A度E边}(\mathcal{E}_{new})$ 
12: 结束循环

```

3.5 实现

我们将MAGMA实现为一个模块化三层架构，旨在实现可扩展性、可伸缩性和部署灵活性。存储层抽象了异构物理后端，为管理类型化记忆图、密集向量索引和稀疏关键词索引提供了统一接口。这种抽象清晰地地将逻辑记忆模型与其物理实现分离，使得能够以最小的工程努力无缝替换存储后端（例如，内存数据结构与生产级图或向量数据库）。

检索层协调核心算法组件，包括记忆构建、多阶段排序和策略引导的图遍历。它由用于情景分割和时间归一化的专用工具模块提供支持，这些模块为下游检索和遍历策略提供结构化信号。应用层管理交互循环、评估框架和提示构建，作为智能体与底层记忆系统之间的接口。

4 实验

我们进行了全面的实验，以评估所提出的 MAGMA 架构在推理有效性和系统属性方面相对于最先进基线模型的表现。

4.1 实验设置

数据集。我们使用两个广泛采用的基准来评估长期对话能力：**(1)LoCoMo(Maharana 等人, 2024):** 包含用于评估长程时间与因果检索的超长对话（平均长度为 9K 词元）。**(2) LongMemEval(Wu 等人,2024):** 一个大规模压力测试基准，其

Table 1: Performance on the LoCoMo benchmark evaluated using the LLM-as-a-Judge metric. Higher scores indicate better performance. LLM model is based on gpt-4o-mini.

Method	Multi-Hop	Temporal	Open-Domain	Single-Hop	Adversarial	Overall
Full Context	0.468	0.562	0.486	0.630	0.205	0.481
A-MEM	0.495	0.474	0.385	0.653	0.616	0.580
MemoryOS	0.552	0.422	0.504	0.674	0.428	0.553
Nemori	0.569	0.649	0.485	0.764	0.325	0.590
MAGMA (ours)	0.528	0.650	0.517	0.776	0.742	0.700

an average context length exceeding 100K tokens, used to evaluate scalability and memory retention stability over extended interaction horizons..

Baselines. We compare MAGMA against four state-of-the-art memory architectures. For fair comparison, all methods employ the same backbone LLMs.

- **Full Context:** Feeds the entire conversation history into the LLM.
- **A-MEM (Xu et al., 2025):** A biological-inspired, self-evolving memory system that dynamically organizes agent experiences.
- **Nemori (Nan et al., 2025):** A graph-based memory utilizing a "predict-calibrate" mechanism for episodic segmentation.
- **MemoryOS(Kang et al., 2025a)** : A semantic-focused memory operating system employing a hierarchical storage strategy.

Metrics. Following standard evaluation protocols, we primarily use the LLM-as-a-Judge score (Zheng et al., 2023) to assess the accuracy of different methods. The detailed evaluation prompt used for the judge model is provided in the appendix. For completeness, we also report token-level F1 and BLEU-1 (Papineni et al., 2002).

4.2 Overall Comparison

This section introduces the accuracy performance comparison between all methods on the LoCoMo benchmark based on LLM-as-a-judge. As shown in Table 1, MAGMA achieves the highest overall judge score of 0.7, substantially outperforming the other baselines: Full Context (0.481), A-MEM (0.58), MemoryOS (0.553) and Nemori (0.59) by relative margins of 18.6% to 45.5%. This result demonstrates that explicitly modeling multi-relational structure enables more accurate long-horizon reasoning than flat or purely semantic memory architectures.

A closer analysis reveals that MAGMA’s advantage is particularly pronounced in reasoning-

intensive settings. In the Temporal category, MAGMA slightly but consistently outperforms others (Judge: 0.650 for MAGMA vs. 0.422 - 0.649 for others), validating the effectiveness of our Temporal Inference Engine in resolving relative temporal expressions into grounded chronological representations. The performance gap further widens under adversarial conditions, where MAGMA attains a judge score of 0.742. This robustness stems from the Adaptive Traversal Policy, which prioritizes causal and entity-consistent paths and avoids semantically similar yet structurally irrelevant distractors that often mislead vector-based retrieval systems. Additional results and analyzes, including case studies and evaluations under alternative metrics, are provided in the appendix.

4.3 Generalization Study

To evaluate generalization under extreme context lengths, we compare MAGMA against prior methods on the LongMemEval benchmark. LongMemEval poses a substantial scalability challenge, with an average context length exceeding 100k tokens, and therefore serves as a rigorous stress test for long-term memory retention and retrieval under strict computational constraints.

As summarized in Table 2, MAGMA achieves the highest average accuracy (61.2%), outperforming both the Full-context baseline (55.0%) and the Nemori system (56.2%). These results indicate that MAGMA generalizes effectively to ultra-long interaction histories while maintaining strong retrieval precision.

At the same time, the results highlight a favorable efficiency–granularity trade-off. Although the Full-context baseline performs strongly on *single-session-assistant* tasks (89.3%), this performance comes at a prohibitive computational cost, requiring over 100k tokens per query. MAGMA achieves competitive accuracy (83.9%) while using only 0.7k–4.2k tokens per query, representing a reduction of more than 95%. This demonstrates that

表1: 使用LLM即法官指标评估的LoCoMo基准测试性能。分数越高表示性能越好。LLM模型基于gpt-4o-mini。

方法	多跳	时序	开放域	单跳	对抗性	总体
完整上下文	0.468	0.562	0.486	0.630	0.205	0.481
A-Mem	0.495	0.474	0.385	0.653	0.616	0.580
MemoryOS	0.552	0.422	0.504	0.674	0.428	0.553
Nemori	0.569	0.649	0.485	0.764	0.325	0.590
MAGMA (我们的方法)	0.528	0.650	0.517	0.776	0.742	0.700

平均上下文长度超过 100K 词元，用于评估在扩展交互范围内的可扩展性和记忆保持稳定性。

基线模型。我们将MAGMA与四种最先进的记忆架构进行比较。为确保公平比较，所有方法均采用相同的大语言模型作为骨干。

- **完整上下文:** 将整个对话历史输入大语言模型。

- **A-Mem(Xu等人,2025年):** 一种受生物学启发的、自演化的记忆系统，能动态组织智能体经验。

- **Nemori(Nan等人, 2025年):** 一种基于图的记忆系统，利用“预测-校准”机制进行情景分割。

- **MemoryOS(Kang等人, 2025a):** 一种专注于语义的记忆操作系统，采用分层存储策略。

指标。遵循标准评估协议，我们主要使用LLM即法官评分 (Zhenget al., 2023) 来评估不同方法的准确性。用于法官模型的详细评估提示词见附录。为求全面，我们也报告词元级 F1分数和BLEU-1分数 (Papineni et al., 2002)。

4.2 总体比较

本节介绍基于LLM即法官，所有方法在LoCoMo基准测试上的准确性性能比较。如表1所示，MAGMA获得了最高的总体法官评分0.7，显著优于其他基线模型：完整上下文（0.481）、A-Mem（0.58）、MemoryOS（0.553）和Nemori（0.59），相对优势幅度在18.6%到45.5%之间。这一结果表明，显式建模多关系结构比扁平或纯语义记忆架构能实现更准确的长程推理。

更深入的分析表明，MAGMA的优势在推理密集型场景中尤为显著。

在时序类别中，MAGMA略微但持续地优于其他方法（Judge评分：MAGMA为0.650，其他方法为0.422 - 0.649），这验证了我们的时序推理引擎在将相对时序表达式解析为基于事实的时间顺序表示方面的有效性。在对抗性条件下，性能差距进一步拉大，此时MAGMA获得了0.742的Judge评分。这种鲁棒性源于自适应遍历策略，该策略优先选择因果和实体一致的路径，并避免了那些语义相似但结构无关的干扰项，这些干扰项常常会误导基于向量的检索系统。其他结果与分析，包括案例研究以及基于替代指标的评估，均在附录中提供。

4.3 泛化性研究

为了评估在极端上下文长度下的泛化能力，我们在LongMemEval基准测试中将MAGMA与先前的方法进行了比较。LongMemEval提出了巨大的可扩展性挑战，其平均上下文长度超过10万个标记，因此可作为在严格计算约束下对长期记忆保持与检索的严格压力测试。

如表2所总结，MAGMA取得了最高的平均准确率（61.2%），优于全上下文基线（55.0%）和Nemori系统（56.2%）。这些结果表明，MAGMA能有效地泛化到超长的交互历史，同时保持强大的检索精度。

与此同时，结果突显了一种有利的效率-粒度权衡。尽管全上下文基线在单会话助手任务上表现强劲（89.3%），但这种性能是以极高的计算成本为代价的，每查询需要超过10万个令牌。MAGMA实现了有竞争力的准确率（83.9%），而每查询仅使用0.7k–4.2k个令牌，这意味着减少了超过95%。这表明

Table 2: Performance comparison on LongMemEval dataset across different question types. We compare our MAGMA method against the Full-context baseline and the Nemori system.

Question Type		Full-context (101K tokens)	Nemori (3.7–4.8K tokens)	MAGMA (0.7–4.2K tokens)
gpt-4o-mini	single-session-preference	6.7%	62.7%	73.3%
	single-session-assistant	89.3%	73.2%	83.9%
	temporal-reasoning	42.1%	43.0%	45.1%
	multi-session	38.3%	51.4%	50.4%
	knowledge-update	78.2%	52.6%	66.7%
	single-session-user	78.6%	77.7%	72.9%
	Average	55.0%	56.2%	61.2%

Table 3: System efficiency comparison with total memory build time (in hours), average token consumption per query (in k tokens), and average query latency (in seconds).

Method	Build Time (h)	Tokens/Query (k)	Latency (s)
Full Context	N/A	8.53	1.74
A-MEM	1.01	2.62	2.26
MemoryOS	0.91	4.76	32.68
Nemori	0.29	3.46	2.59
MAGMA	0.39	3.37	1.47

MAGMA effectively compresses long interaction histories into compact, reasoning-dense subgraphs, preserving essential information while substantially reducing inference-time overhead.

4.4 System Efficiency Analysis

To evaluate the system efficiency of MAGMA, two metrics are focused: (1) memory build time (the time required to construct the memory graph) and (2) token cost (the average tokens processed per query).

Table 3 reports the comparative results. While A-MEM achieves the lowest token consumption (2.62k) due to its aggressive summarization, it sacrifices reasoning depth (see Table 1). In contrast, MAGMA achieves the lowest query latency (1.47s) about 40% faster than the next best retrieval baseline (A-MEM) while maintaining a competitive token cost (3.37k). This efficiency stems from our Adaptive Traversal Policy, which prunes irrelevant subgraphs early, and the dual-stream architecture that offloads complex indexing to the background.

4.5 Ablation Study

In this subsection, we conduct a systematic ablation study to assess the contribution of individual components in MAGMA. By selectively disabling edge types and traversal mechanisms, we isolate the sources of its reasoning capability. The results

Table 4: Breakdown analysis on the performance impact of different schemes in MAGMA.

MAGMA schemes	Judge	F1	BLEU-1
w/o Adaptive Policy	0.637	0.413	0.357
w/o Causal Links	0.644	0.439	0.354
w/o Temporal Backbone	0.647	0.438	0.349
w/o Entity Links	0.666	0.451	0.363
MAGMA (Full)	0.700	0.467	0.378

in Table 4 reveal three main findings.

First, removing the Adaptive Policy results in the largest performance drop, with the Judge score decreasing from 0.700 to 0.637. This confirms that intent-aware routing is critical: without it, retrieval degenerates into a static graph walk that introduces structurally irrelevant information and degrades reasoning quality. Second, removing either Causal Links or the Temporal Backbone leads to comparable and substantial performance losses (0.644 and 0.647, respectively), indicating that causal structure and temporal ordering provide complementary, non-substitutable axes of reasoning. Finally, removing Entity Links causes a smaller but consistent decline (0.700 to 0.666), highlighting their role in maintaining entity permanence and reducing hallucinations in entity-centric queries.

To further isolate the contribution of each relation type, we additionally conducted a single-graph-only ablation on LoCoMo, shown in Table 5. The results are consistent with the leave-one-out findings in Table 4. Among single-graph variants, *Causal Only* achieves the highest overall score (0.590), suggesting that causal relations provide strong logical filtering and robustness against distractor noise. *Temporal Only* performs best on temporal questions (0.620), confirming that explicit temporal structure is particularly important for sequential reasoning. In contrast, *Entity Only* obtains the lowest overall score (0.531): while it remains

表2: 在LongMemEval数据集上针对不同问题类型的性能比较。我们将MAGMA方法与全上下文基线及Nemori系统进行对比。

问题类型		全上下文 (101K tokens)	Nemori (3.7–4.8K 词元)	MAGMA (0.7–4.2K 词元)
gpt-4o-mini	单会话偏好	6.7%	62.7%	73.3%
	单会话助手	89.3%	73.2%	83.9%
	时间推理	42.1%	43.0%	45.1%
	多会话	38.3%	51.4%	50.4%
	知识更新	78.2%	52.6%	66.7%
	单会话用户	78.6%	77.7%	72.9%
	平均值	55.0%	56.2%	61.2%

表 3: 系统效率对比，包含总内存构建时间（单位：小时）、平均每查询令牌消耗（单位：千令牌）以及平均查询延迟（单位：秒）。

方法	构建时间(小时)	每查询令牌数 (千)	延迟 (秒)
完整上下文	N/A	8.53	1.74
A-Mem	1.01	2.62	2.26
MemoryOS	0.91	4.76	32.68
Nemori	0.29	3.46	2.59
MAGMA	0.39	3.37	1.47

MAGMA有效地将长交互历史压缩成紧凑、推理密集的子图，在保留关键信息的同时，大幅降低了推理时间开销。

4.4 系统效率分析

为评估 MAGMA 的系统效率，我们重点关注两个指标：(1) 内存构建时间（构建记忆图所需的时间）和 (2) 令牌成本（每查询平均处理令牌数）。

表3报告了对比结果。虽然 A-Mem 因其激进摘要实现了最低的令牌消耗（2.62k），但它牺牲了推理深度（见表1）。相比之下，MAGMA 实现了最低的查询延迟（1.47秒），比次优的检索基线（A-Mem）快约40%，同时保持了有竞争力的令牌成本（3.37k）。这种效率源于我们的自适应遍历策略（它能及早剪除不相关子图）以及将复杂索引卸载到后台的双流架构。

4.5 消融研究

在本小节中，我们进行了一项系统的消融研究，以评估MAGMA中独立组件的贡献。通过选择性禁用边类型和遍历机制，我们分离出其推理能力的来源。结果

表 4: MAGMA 中不同方案对性能影响的细分分析。

MAGMA方案	Judge评分	F1	BLEU-1分数
无自适应策略	0.637	0.413	0.357
无因果链接	0.644	0.439	0.354
无时间骨干	0.647	0.438	0.349
无实体链接	0.666	0.451	0.363
MAGMA (完整版)	0.700	0.467	0.378

表4 揭示了三个主要发现。

首先，移除自适应策略会导致性能下降最大，Judge评分从0.700降至0.637。这证实了意图感知路由至关重要：没有它，检索会退化为静态图遍历，从而引入结构上不相关的信息并降低推理质量。其次，移除因果链接或时间骨干都会导致相当且显著的性能损失（分别为0.644和0.647），这表明因果结构和时序排序提供了互补且不可替代的推理维度。最后，移除实体链接会导致较小但一致的性能下降（从0.700到0.666），这突显了它们在维持实体持久性和减少以实体为中心的查询中的幻觉方面的作用。

为了进一步分离每种关系类型的贡献，我们在LoCoMo上额外进行了单图消融实验，结果如表5所示。这些结果与表4中的留一法发现一致。在单图变体中，仅因果获得了最高的总体得分（0.590），这表明因果关系提供了强大的逻辑过滤能力，并对干扰噪声具有鲁棒性。仅时序在时序问题上表现最佳（0.620），证实了显式的时间结构对于顺序推理尤为重要。相比之下，仅实体获得了最低的总体得分（0.531）：尽管它在多跳

Table 5: Single-graph ablation study on LoCoMo.

Graph Configuration	Multi-Hop	Temporal	Open-Domain	Single-Hop	Adversarial	Overall
Causal Only	0.470	0.460	0.430	0.650	0.680	0.590
Temporal Only	0.440	0.620	0.450	0.650	0.520	0.577
Entity Only	0.485	0.420	0.460	0.640	0.450	0.531
Full MAGMA	0.528	0.650	0.517	0.776	0.742	0.700

helpful for concept bridging in multi-hop reasoning, it lacks both timeline awareness and logical filtering, which explains why removing entity links causes the smallest drop in the full-system ablation.

Overall, these results show that no single relation type is sufficient to recover MAGMA’s full reasoning capability, as all single-graph variants remain below 0.60 overall. By explicitly decoupling causal, temporal, and entity relations and combining them with adaptive traversal, MAGMA leverages their complementary strengths to achieve the best overall performance.

5 Conclusion

We introduced MAGMA, a multi-graph agentic memory architecture that models semantic, temporal, causal, and entity relations within a unified yet disentangled memory substrate. By formulating retrieval as a policy-guided graph traversal and decoupling memory ingestion from asynchronous structural consolidation, MAGMA enables effective long-horizon reasoning while maintaining low inference-time latency. Empirical results on LoCoMo and LongMemEval demonstrate that MAGMA consistently outperforms state-of-the-art memory systems while achieving substantial efficiency gains under ultra-long contexts.

6 Limitations

While MAGMA demonstrates strong empirical performance, it has several limitations. First, the quality of the constructed memory graph depends on the reasoning fidelity of the underlying Large Language Models used during asynchronous consolidation. This dependency is a shared limitation of agentic memory systems that rely on LLM-based structural inference, as they are susceptible to extraction errors and hallucinations (Pan et al., 2024; Xi et al., 2025; Wadhwa et al., 2023). Although MAGMA employs structured prompts and conservative inference thresholds to reduce spurious links, erroneous or missing relations may still arise and propagate to downstream retrieval. Nevertheless,

our experimental results indicate that, even under these constraints, agentic memory systems such as MAGMA substantially outperform traditional baselines, including full-context approaches, in long-horizon reasoning tasks.

Second, multi-graph substrate may introduce additional storage and engineering complexity compared to flat, vector-only memory systems. Maintaining multiple relational views and dual-stream processing incurs a little higher implementation and memory overhead, which may limit applicability in highly resource-constrained environments.

Finally, most existing agentic memory systems, including MAGMA, are primarily evaluated on long-context conversational and agentic benchmarks such as LoCoMo and LongMemEval. While these benchmarks effectively stress temporal and causal reasoning, they do not cover the full range of settings in which agentic memory may be required(Hu et al., 2025; Jiang et al., 2026). Extending MAGMA to other scenarios, such as multi-modal agents or environments with heterogeneous observation streams, may require additional adaptation and calibration. Addressing these broader evaluation settings remains an important research direction for future work.

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References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.
- Iz Beltagy, Matthew E Peters, and Arman Cohan. 2020a. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.

表5: LoCoMo上的单图消融研究。

图配置	多跳	时序	开放域	单跳	对抗性	总体
仅因果	0.470	0.460	0.430	0.650	0.680	0.590
仅时序	0.440	0.620	0.450	0.650	0.520	0.577
仅实体	0.485	0.420	0.460	0.640	0.450	0.531
完整MAGMA	0.528	0.650	0.517	0.776	0.742	0.700

推理中的概念桥接方面仍有帮助，但它既缺乏时间线感知能力，也缺乏逻辑过滤能力，这解释了为何在全系统消融中移除实体链接导致的下降幅度最小。

总体而言，这些结果表明，没有任何单一关系类型足以恢复MAGMA的全部推理能力，因为所有单图变体的总体得分均低于0.60。通过显式解耦因果、时序和实体关系，并将它们与自适应遍历相结合，MAGMA利用了它们的互补优势，从而实现了最佳的总体性能。

5 结论

我们提出了MAGMA，一种多图智能体记忆架构，它在统一且解耦的记忆基底中对语义、时序、因果和实体关系进行建模。通过将检索表述为策略引导的图遍历，并将记忆摄取与异步结构巩固解耦，MAGMA在保持低推理时延的同时，实现了有效的长视野推理。在LoCoMo和LongMemEval上的实证结果表明，MAGMA在超长上下文下持续优于最先进的记忆系统，同时实现了显著的效率提升。

6 局限性

尽管MAGMA展现出强大的实证性能，但它也存在一些局限性。首先，所构建的记忆图的质量取决于异步整合过程中所使用的基础大语言模型的推理保真度。这种依赖性是基于大语言模型结构推断的智能体记忆系统共有的局限，因为它们容易受到提取错误和幻觉的影响（Pan等人,2024; Xi等人, 2025年; Wadhwa等人, 2023）。尽管MAGMA采用了结构化提示词和保守的推理阈值来减少虚假链接，但错误或缺失的关系仍可能出现，并传播到下游检索中。尽管如此，

我们的实验结果表明，即使在这些限制下，像MAGMA这样的智能体记忆系统在长程推理任务中的表现也显著优于包括全上下文方法在内的传统基线。

其次，与扁平、纯向量记忆系统相比，多图基底可能会引入额外的存储和工程复杂性。维护多个关系视图和双流处理会产生略高的实现与内存开销，这可能会限制其在高度资源受限环境中的适用性。

最后，包括MAGMA在内的大多数现有智能体记忆系统，主要是在长上下文对话和智能体基准测试（如LoCoMo和LongMemEval）上进行评估的。虽然这些基准测试有效地考验了时序和因果推理能力，但并未涵盖可能需要智能体记忆的全部场景 (Hu等人, 2025年; Jiang等人, 2026)。将MAGMA扩展到其他场景，例如多模态智能体或具有异构观测流的环境，可能需要额外的适配和校准。解决这些更广泛的评估设置，仍然是未来工作的重要研究方向。

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参考文献

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, 及另外1人。2023. GPT-4技术报告. *arXiv预印本 arXiv:2303.08774*.
- Iz Beltagy、Matthew E Peters 和 ArmanCohan。2020a. Longformer: 长文档Transformer. *arXiv预印本 arXiv:2004.05150*.

Iz Beltagy, Matthew E Peters, and Arman Cohan. 2020b. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*.

Gordon V. Cormack, Charles L A Clarke, and Stefan Buettcher. 2009. [Reciprocal rank fusion outperforms condorcet and individual rank learning methods](#). In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR ’09, page 758–759, New York, NY, USA. Association for Computing Machinery.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitanasky, Robert Osazuwa Ness, and Jonathan Larson. 2024a. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitanasky, Robert Osazuwa Ness, and Jonathan Larson. 2024b. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. 2025. From rag to memory: Non-parametric continual learning for large language models. *arXiv preprint arXiv:2502.14802*.

Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, Boyang Liu, Fangyi Zhu, Jiahang Lin, Honglin Guo, Shihan Dou, Zhiheng Xi, and 1 others. 2025. Memory in the age of ai agents. *arXiv preprint arXiv:2512.13564*.

Le Huang, Hengzhi Lan, Zijun Sun, Chuan Shi, and Ting Bai. 2024. Emotional rag: Enhancing role-playing agents through emotional retrieval. In *2024 IEEE International Conference on Knowledge Graph (ICKG)*, pages 120–127. IEEE.

Dongming Jiang, Yi Li, Songtao Wei, Jinxin Yang, Ayushi Kishore, Alysa Zhao, Dingyi Kang, Xu Hu, Feng Chen, Qiannan Li, and 1 others. 2026. Anatomy of agentic memory: Taxonomy and empirical analysis of evaluation and system limitations. *arXiv preprint arXiv:2602.19320*.

Wenqi Jiang, Suvinay Subramanian, Cat Graves, Gustavo Alonso, Amir Yazdanbakhsh, and Vidushi Dadu. 2025. Rago: Systematic performance optimization

for retrieval-augmented generation serving. In *Proceedings of the 52nd Annual International Symposium on Computer Architecture*, pages 974–989.

Ziyan Jiang, Xueguang Ma, and Wenhu Chen. 2024. Longrag: Enhancing retrieval-augmented generation with long-context llms. *arXiv preprint arXiv:2406.15319*.

Zhijing Jin, Yuen Chen, Felix Leeb, Luigi Gresele, Ojasv Kamal, Zhiheng Lyu, Kevin Blin, Fernando Gonzalez Adauto, Max Kleiman-Weiner, Mrinmaya Sachan, and 1 others. 2023. Cladder: Assessing causal reasoning in language models. *Advances in Neural Information Processing Systems*, 36:31038–31065.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2019. Billion-scale similarity search with gpus. *IEEE Transactions on Big Data*, 7(3):535–547.

Jiazheng Kang, Mingming Ji, Zhe Zhao, and Ting Bai. 2025a. Memory os of ai agent. *arXiv preprint arXiv:2506.06326*.

Jiazheng Kang, Mingming Ji, Zhe Zhao, and Ting Bai. 2025b. Memory os of ai agent. *arXiv preprint arXiv:2506.06326*.

Jikun Kang, Wenqi Wu, Filippos Christianos, Alex J Chan, Fraser Greenlee, George Thomas, Marvin Purtorab, and Andy Toulis. 2025c. Lm2: Large memory models. *arXiv preprint arXiv:2502.06049*.

Urvashi Khandelwal, He He, Peng Qi, and Dan Jurafsky. 2018. Sharp nearby, fuzzy far away: How neural language models use context. *arXiv preprint arXiv:1805.04623*.

Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. 2023. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research*.

Dharshan Kumaran, Demis Hassabis, and James L McClelland. 2016. What learning systems do intelligent agents need? complementary learning systems theory updated. *Trends in cognitive sciences*, 20(7):512–534.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474.

Yi Li, Lianjie Cao, Faraz Ahmed, Puneet Sharma, and Bingzhe Li. 2026. Hippocampus: An efficient and scalable memory module for agentic ai. *arXiv preprint arXiv:2602.13594*.

Zhiyu Li, Shichao Song, Chenyang Xi, Hanyu Wang, Chen Tang, and 1 others. 2025. Memos: A memory os for ai system. *arXiv preprint arXiv:2507.03724*.

Iz Beltagy, Matthew E Peters 和 ArmanCohan。2020b. Longformer: 长文档Transformer。 *arXiv*预印本 *arXiv:2004.05150*。

Tom Brown、 Benjamin Mann、 Nick Ryder、 Melanie Subbiah、 Jared D Kaplan、 Prafulla Dhariwal、 Arvind Neelakantan、 Pranav Shyam、 Girish Sastry、 A manda Askell 及另外 1 人。2020。语言模型是少样本学习器。 *神经信息处理系统进展*， 33: 1877–1901。

Prateek Chhikara、 Dev Khant、 Saket Aryan、 Taranjeet Singh 和 Deshraj Yadav。2025年。Mem0: 构建具备可扩展长期记忆的生产就绪AI智能体。 *arXiv*预印本 *arXiv:2504.19413*。

Gordon V. Cormack、 Charles L A Clarke 和 Stefan Buettcher。2009。 [互逆排序融合优于孔多塞和个体排序学习方法](#)。收录于第32届国际 *ACM SIGIR*信息检索研究与发展会议论文集， SIGIR ’ 09， 第758–759页， 美国纽约州纽约市。计算机协会。

Darren Edge、 Ha Trinh、 Newman Cheng、 Joshua Bradley、 Alex Chao、 Apurva Mody、 Steven Truitt、 Dasha Metropolitanasky、 Robert Osazuwa Ness 和 Jonathan Larson。2024a。从局部到全局：一种面向查询聚焦式摘要的图RAG方法。 *arXiv*预印本 *arXiv:2404.16130*。

Darren Edge、 Ha Trinh、 Newman Cheng、 Joshua Bradley、 Alex Chao、 Apurva Mody、 Steven Truitt、 Dasha Metropolitanasky、 Robert Osazuwa Ness 和 Jonathan Larson。2024b。从局部到全局：一种面向查询聚焦式摘要的图RAG方法。 *arXiv*预印本 *arXiv:2404.16130*。 Bernal Jiménez Gutiérrez、 Yiheng Shu、 Weijian Qi、 Sizhe Zhou 和 Yu Su。2025年。从RAG到记忆：面向大语言模型的非参数持续学习。 *arXiv*预印本 *arXiv:2502.14802*。 Yuyang Hu、 Shichun Liu、 Yanwei Yue、 Guibin Zhang、 Boyang Liu、 Fangyi Zhu、 Jiahang Lin、 Honglin Guo、 Shihan Dou、 Zhiheng Xi 及另外1人。2025年。AI智能体时代的记忆。 *arXiv*预印本 *arXiv:2512.13564*。 Le Huang、 Hengzhi Lan、 Zijun Sun、 Chuan Shi 和 Ting Bai。2024年。情感RAG：通过情感检索增强角色扮演智能体。载于2024年 *IEEE*知识图谱国际会议 (*ICKG*)， 第120–127页。IEEE。 Dongming Jiang、 Yi Li、 Songtao Wei、 Jinxin Yang、 Ayushi Kishore、 Alysa Zhao、 Dingyi Kang、 Xu Hu、 Feng Chen、 Qiannan Li 及另外1人。2026。能动记忆剖析：评估与系统局限性的分类法及实证分析。 *arXiv*预印本 *arXiv:2602.19320*。 Wenqi Jiang、 Suvinay Subramanian、 Cat Graves、 Gustavo Alonso、 Amir Yazdanbakhsh 和 Vidushi Dadu。2025年。RAGO：系统性性能优化

用于检索增强生成服务。载于第52届计算机体系结构国际年会论文集， 第974–989页。

ZiyanJiang、 Xueguang Ma 和 Wenhu Chen。2024。Longrag： 利用长上下文大语言模型增强检索增强生成。 *arXiv*预印本 *arXiv:2406.15319*。

Zhijing Jin、 Yuen Chen、 Felix Leeb、 Luigi Gresele、 Ojasv Kamal、 Zhiheng Lyu、 Kevin Blin、 Fernando Gonzalez Adauto、 Max Kleiman-Weiner、 Mrinmaya Sachan 及另外 1 人。2023。Cladder： 评估语言模型中的因果推理。 *神经信息处理系统进展*， 36: 31038–31065。

Jeff Johnson、 Matthijs Douze、 和 Hervé Jégou。2019。十亿规模相似性搜索与 GPU。 *IEEE* 大数据汇刊， 7(3):535–547。

JiazhengKang、 MingmingJi、 Zhe Zhao、 和 Ting Bai。2025a。AI 代理的内存操作系统。 *arXiv*预印本 *arXiv:2506.06326*。

JiazhengKang、 MingmingJi、 Zhe Zhao、 和 Ting Bai。2025b。AI 代理的内存操作系统。 *arXiv*预印本 *arXiv:2506.06326*。

Jikun Kang、 Wenqi Wu、 Filippos Christianos、 Alex J Chan、 Fraser Greenlee、 George Thomas、 Marvin Purtorab、 and Andy Toulis。2025c。LM2： 大型记忆模型。 *arXiv*预印本 *arXiv:2502.06049*。 Urvashi Khandelwal、 He He、 Peng Qi、 and Dan Jurafsky。2018。近处清晰，远处模糊：神经语言模型如何使用上下文。 *arXiv*预印本 *arXiv:1805.04623*。 Emre Kiciman、 Robert Ness、 Amit Sharma、 and Chenhao Tan。2023。因果推理与大语言模型：开启因果性的新前沿。机器学习研究汇刊。Dharshan Kumaran、 Demis Hassabis、 and James L McClelland。2016。智能体需要什么样的学习系统？互补学习系统理论更新。认知科学趋势， 20(7):512–534。 Patrick Lewis、 Ethan Perez、 Aleksandra Piktus、 Fabio Petroni、 Vladimir Karpukhin、 Naman Goyal、 Heinrich Küttler、 Mike Lewis、 Wen-tau Yih、 Tim Rocktäschel、 and 1 others。2020。面向知识密集型自然语言处理任务的检索增强生成。 *神经信息处理系统进展*， 33:9459–9474。 Yi Li、 Lianjie Cao、 Faraz Ahmed、 Puneet Sharma、 and Bingzhe Li。2026。Hippocampus： 一个高效且可扩展的代理型AI记忆模块。 *arXiv*预印本 *arXiv:2602.13594*。 Zhiyu Li、 Shichao Song、 Chenyang Xi、 Hanyu Wang、 Chen Tang、 and 1 others。2025。Memos： 一个用于AI系统的内存操作系统。 *arXiv*预印本 *arXiv:2507.03724*。

Shuhang Lin, Zhencan Peng, Lingyao Li, Xiao Lin, Xi Zhu, and Yongfeng Zhang. 2025. Cache mechanism for agent rag systems. *arXiv preprint arXiv:2511.02919*.

Lei Liu, Xiaoyan Yang, Yue Shen, Binbin Hu, Zhiqiang Zhang, Jinjie Gu, and Guannan Zhang. 2023. Think-in-memory: Recalling and post-thinking enable llms with long-term memory. *arXiv preprint arXiv:2311.08719*.

Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17753*.

Jiayan Nan, Wenquan Ma, Wenlong Wu, and Yize Chen. 2025. Nemori: Self-organizing agent memory inspired by cognitive science. *arXiv preprint arXiv:2508.03341*.

Charles Packer, Vivian Fang, Shishir_G Patil, Kevin Lin, Sarah Wooders, and Joseph_E Gonzalez. 2023. Memgpt: Towards llms as operating systems.

Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Jipu Wang, and Xindong Wu. 2024. Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering*, 36(7):3580–3599.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. **Bleu: a method for automatic evaluation of machine translation**. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.

Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pages 1–22.

Ofir Press, Noah A Smith, and Mike Lewis. 2021a. Train short, test long: Attention with linear biases enables input length extrapolation. *arXiv preprint arXiv:2108.12409*.

Ofir Press, Noah A Smith, and Mike Lewis. 2021b. Train short, test long: Attention with linear biases enables input length extrapolation. *arXiv preprint arXiv:2108.12409*.

Hongjin Qian, Zheng Liu, Peitian Zhang, Kelong Mao, Defu Lian, Zhicheng Dou, and Tiejun Huang. 2025. Memorag: Boosting long context processing with

global memory-enhanced retrieval augmentation. In *Proceedings of the ACM on Web Conference 2025*, pages 2366–2377.

Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. Zep: a temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*.

Somin Wadhwa, Silvio Amir, and Byron C Wallace. 2023. Revisiting relation extraction in the era of large language models. In *Proceedings of the conference. association for computational linguistics. meeting*, volume 2023, page 15566.

Yu Wang and Xi Chen. 2025. Mirix: Multi-agent memory system for llm-based agents. *arXiv preprint arXiv:2507.07957*.

Zheng Wang, Shu Teo, Jieer Ouyang, Yongjun Xu, and Wei Shi. 2024. M-rag: Reinforcing large language model performance through retrieval-augmented generation with multiple partitions. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1966–1978.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*.

Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwending, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, and 1 others. 2025. The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, 68(2):121101.

Shijia Xu, Zhou Wu, Xiaolong Jia, Yu Wang, Kai Liu, and April Xiaowen Dong. 2026. **Self-correcting rag: Enhancing faithfulness via mmkp context selection and nli-guided mcts**. *Preprint*, arXiv:2604.10734.

Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-mem: Agentic memory for llm agents. *arXiv preprint arXiv:2502.12110*.

Yiming Zeng, Wanhao Yu, Zexin Li, Tao Ren, Yu Ma, Jinghan Cao, Xiyan Chen, and Tingting Yu. 2025. Bridging the editing gap in llms: Fineedit for precise and targeted text modifications. *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 2193–2206.

Zhuosheng Zhang, Yao Yao, Aston Zhang, Xiangru Tang, Xinbei Ma, Zhiwei He, Yiming Wang, Mark Gerstein, Rui Wang, Gongshen Liu, and 1 others.

Shuhang Lin, Zhencan Peng, Lingyao Li, Xiao Lin, Xi Zhu, and Yongfeng Zhang. 2025. 智能体RAG系统的缓存机制. *arXiv预印本 arXiv:2511.02919*.

Lei Liu, Xiaoyan Yang, Yue Shen, Binbin Hu, Zhiqiang Zhang, Jinjie Gu 和 Guannan Zhang. 2023. 记忆内思考：召回率与后思考赋能大语言模型长期记忆. *arXiv预印本 arXiv:2311.08719*.

Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. 迷失在中间：语言模型如何使用长上下文。 计算语言学协会汇刊, 12:157–173.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri 和 Yuwei Fang. 2024年. 评估大语言模型智能体的超长期对话记忆. *arXiv预印本 arXiv:2402.17753*.

Jiayan Nan, Wenquan Ma, Wenlong Wu 和 Yize Chen. 2025年. Nemori: 受认知科学启发的自组织智能体记忆. *arXiv预印本 arXiv:2508.03341*.

Charles Packer, Vivian Fang, Shishir_G Patil, Kevin Lin、 莎拉 Wooders 和 Joseph_E Gonzalez. 2023年. Memgpt: 迈向将大语言模型作为操作系统。

Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Ji- apu Wang, and Xindong Wu. 2024. 统一大语言模型与知识图谱：路线图。

IEEE Transactions on Knowledge and Data Engineering, 36(7):3580–3599. Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. **Bleu: 一种机器翻译自动评估方法**。载于计算语言学协会第40届年会论文集，第311–318页，美国宾夕法尼亚州费城。计算语言学协会。 Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. 生成式智能体：人类行为的交互式模拟。载于第36届ACM用户界面软件与技术年会论文集，第1–22页。 Ofir Press, Noah A Smith, and Mike Lewis.

2021a. 短训练，长测试：线性偏置注意力机制实现输入长度外推. *arXiv预印本 arXiv:2108.12409*. Ofir Press, Noah A Smith, and Mike Lewis. 2021b. 短训练，长测试：线性偏置注意力机制实现输入长度外推. *arXiv预印本 arXiv:2108.12409*. Hongjin Qian, Zheng Liu, Peitian Zhang, Kelong Mao, Defu Lian, Zhicheng Dou, and Tiejun Huang. 2025. Memorag：通过

全局记忆增强的检索增强生成提升长上下文处理。载于ACM 网络会议 2025 论文集，第2366–2377页。

Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan 和 Daniel Chalef. 2025年. Zep：一种用于代理记忆的时序知识图谱架构. *arXiv预印本 arXiv:2501.13956*.

Somin Wadhwa, Silvio Amir 和 Byron C Wallace. 2023年. 在大语言模型时代重新审视关系抽取。载于计算语言学协会会议论文集，第2023卷，第15566页。

Yu Wang 和 Xi Chen. 2025年. Mirix：面向基于大语言模型的代理的多智能体记忆系统. *arXiv预印本 arXiv:2507.07957*.

Zheng Wang, Shu Teo, Jieer Ouyang, Yongjun Xu 和 Wei Shi. 2024年. M-rag：通过多分区检索增强生成强化大语言模型性能。载于计算语言学协会第62届年会论文集（第一卷：长论文），第1966–1978页。

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou 及另外1人。2022年. 思维链提示激发大语言模型中的推理。神经信息处理系统进展，35：24824–24837。

DiWu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv预印本 arXiv:2410.10813*. Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwending, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, and 1 others. 2025. The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, 68(2):121101. Shijia Xu, Zhou Wu, Xiaolong Jia, Yu Wang, Kai Liu, and April Xiaowen Dong. 2026. **Self-correcting rag: Enhancing faithfulness via mmkp context selection and nli-guided mcts**. 预印本, arXiv:2604.10734. Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-mem: Agentic memory for llm agents. *arXiv预印本 arXiv:2502.12110*. Yiming Zeng, Wanhao

Yu, Zexin Li, Tao Ren, Yu Ma, Jinghan Cao, Xiyan Chen, and Tingting Yu. 2025. Bridging the editing gap in llms: Fineedit for precise and targeted text modifications. 计算语言学协会发现: *EMNLP 2025*, pages 2193–2206. Zhuosheng Zhang, Yao Yao, Aston Zhang, Xiangru Tang, Xinbei Ma, Zhiwei He, Yiming Wang, Mark Gerstein, Rui Wang, Gongshen Liu, and 1 others.

2025. Igniting language intelligence: The hitchhiker’s guide from chain-of-thought reasoning to language agents. *ACM Computing Surveys*, 57(8):1–39.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, and 1 others. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. Memorybank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19724–19731.

A Related Work

Following the framing in main text, we organize related work along the same progression: from context window extension to retrieval augmented generation (RAG) and finally to memory augmented generation (MAG), and then discuss structured/graph memories and causal reasoning, which are central to long-horizon agentic interactions.

Context-window Extension. A direct line of work extends the effective context length of Transformers by modifying attention or positional extrapolation. Longformer (Beltagy et al., 2020b) introduces sparse attention patterns to scale to long documents, reducing quadratic cost while retaining locality and selected global connectivity. ALiBi (Press et al., 2021b) (Attention with Linear Biases) enables length extrapolation by injecting distance-aware linear biases into attention scores, improving robustness when testing on longer sequences than those seen in training. Recent efforts also add explicit memory modules or hybrid mechanisms to push beyond pure attention-window scaling. For example, LM2 (Kang et al., 2025c) proposes a decoder-only architecture augmented with an auxiliary memory to mitigate long-context limitations. MemoRAG (Qian et al., 2025) similarly emphasizes global-memory-enhanced retrieval to boost long-context processing when raw context is insufficient or inefficient. While these approaches improve long-range coverage, they do not, by themselves, address the continual, evolving, and write-back nature of agent memory required for multi-session interactions.

Retrieval Augmented Generation. RAG (Lewis et al., 2020) augments an LLM with external retrieval over a fixed corpus, classically retrieving supporting passages and conditioning generation

on them. Subsequent work explores better integration with long-context models and more scalable retrieval pipelines. LongRAG (Jiang et al., 2024) studies how to exploit long-context LLMs together with retrieval, improving the ability to incorporate larger retrieved evidence sets. Other systems focus on structuring the retrieved memory space or optimizing the RAG serving stack: M-RAG (Wang et al., 2024) uses multiple partitions to encourage fine-grained retrieval focus, while RAGO (Jiang et al., 2025) provides a systematic framework for performance optimization in RAG serving. Furthermore, to improve the reliability of retrieved context, approaches like Self-Correcting RAG (Xu et al., 2026) enhance generation faithfulness via selective context filtering and inference-guided search. However, standard RAG typically assumes a static knowledge base. In contrast, agentic settings require memory that is continuously updated (the feedback loop described in the main text). This motivates the shift to MAG systems, where memory is dynamic and evolves with interaction histories.

Memory Augmented Generation and Agent Memory Systems. MAG systems maintain and update an external memory over time, enabling agents to accumulate knowledge, preserve identity, and remain coherent across sessions. Early and representative directions include memory construction and write-back strategies for long-term agent behavior, such as MemoryBank (Zhong et al., 2024) and generative agents style architectures that emphasize persistent profiles and evolving state grounded in past interactions (Nan et al., 2025; Maharana et al., 2024; Zeng et al., 2025). A growing body of work adopts systems metaphors and designs: MemGPT (Packer et al., 2023) frames LLM agents with an operating-system-like memory hierarchy, emphasizing paging and controlled context management. More recent memory OS systems propose explicit storage hierarchies, efficient memory modules, and controllers (e.g., MemoryOS (Kang et al., 2025b), MemOS (Li et al., 2025), and Hippocampus (Li et al., 2026)) to manage persistence, scalability, and retrieval policies at scale. In addition, practical agent-memory stacks (e.g., Zep (Rasmussen et al., 2025)) offer temporal knowledge-graph-based memory services aimed at real-world deployment constraints.

Structured memory: chains-of-thought and graph-based representations. Beyond flat text buffers or vector stores, several methods explicitly structure memory to support reasoning. Think-in-

2025. Igniting language intelligence: The hitchhiker’s guide from chain-of-thought reasoning to language agents. *ACM计算概览*, 57(8):1–39.

郑连民、Wei-LinChiang、盛颖、庄思源、吴章昊、庄永浩、林子、李卓翰、李大成、邢波，以及另外1人。2023。使用MT-Bench和Chatbot Arena评判LLM即法官。《神经信息处理系统进展》，36：46595–46623。

钟万钧、郭良宏、高琪琪、叶赫、王彦霖。2024。MemoryBank：为大型语言模型增强长期记忆。载于《AAAI人工智能会议论文集》，第38卷，第19724–19731页。

相关工作

遵循正文中的框架，我们按照相同的脉络组织相关工作：从上下文窗口扩展，到检索增强生成（RAG），再到记忆增强生成（MAG），最后讨论对长视野智能体交互至关重要的结构化/图记忆与因果推理。

上下文窗口扩展。一项直接的研究方向通过修改注意力机制或位置外推来扩展Transformer的有效上下文长度。Longformer (Beltagy et al., 2020b) 引入了稀疏注意力模式以扩展到长文档，在保留局部性和选定的全局连接性的同时降低了二次方成本。ALiBi (Press et al., 2021b)（线性偏置注意力）通过向注意力分数中注入距离感知的线性偏置来实现长度外推，从而提高了在比训练所见更长的序列上进行测试时的鲁棒性。近期的研究也增加了显式的记忆模块或混合机制，以超越纯粹的注意力窗口缩放。例如，LM2 (Kang et al., 2025c) 提出了一种仅解码器架构，并通过辅助记忆进行增强，以缓解长上下文限制。MemoRAG (Qian et al., 2025) 同样强调通过全局记忆增强检索来提升长上下文处理能力，尤其是在原始上下文不足或效率低下时。虽然这些方法改善了长距离覆盖能力，但它们本身并未解决多会话交互所需的、持续演进且可回写的智能体记忆特性。

检索增强生成。RAG (Lewis et al., 2020) 通过在一个固定语料库上进行外部检索来增强大语言模型，经典做法是检索支持性段落并基于此进行生成

。后续工作探索了与长上下文模型更好的集成以及更具可扩展性的检索管道。LongRAG (Jiang et al., 2024) 研究了如何结合利用长上下文大语言模型与检索，提升整合更大检索到的证据集的能力。其他系统专注于构建检索到的记忆空间或优化RAG服务栈：M-RAG (Wang et al., 2024) 使用多分区来鼓励细粒度检索聚焦，而RAGO (Jiang et al., 2025年) 为RAG服务中的性能优化提供了一个系统化框架。此外，为了提高检索到的上下文的可靠性，像Self-Correcting RAG (Xu et al., 2026) 这类方法通过选择性上下文过滤和推理引导搜索来增强生成忠实度。然而，标准RAG通常假设一个静态知识库。相比之下，智能体设置需要持续更新的记忆（如正文中描述的反馈循环）。这促使了向MAG系统的转变，其中的记忆是动态的，并随着交互历史而演进。

记忆增强生成与智能体记忆系统。MAG系统随时间维护和更新外部记忆，使智能体能够积累知识、保持身份一致性，并在多个会话中保持连贯性。早期且具有代表性的研究方向包括针对长期智能体行为的记忆构建与回写策略，例如MemoryBank (Zhong et al., 2024)，以及强调基于过往交互建立持久化角色档案和演化状态的生成式智能体风格架构 (Nan et al., 2025; Maharana et al., 2024; Zeng et al., 2025)。越来越多的研究采用系统隐喻和设计：MemGPT (Packer et al., 2023) 为LLM智能体构建了类似操作系统的记忆层次结构，强调分页和受控的上下文管理。更近期的内存操作系统提出了显式的存储层次、高效的记忆模块和控制器（例如，MemoryOS (Kang et al., 2025b)、MemOS (Li et al., 2025) 和 Hippocampus (Li et al., 2026))，以大规模管理持久性、可扩展性和检索策略。此外，实用的智能体-记忆栈（例如 Zep (Rasmussen et al., 2025)) 提供了基于时序知识图谱的记忆服务，旨在满足现实世界部署的约束条件。

结构化记忆：思维链与基于图的表示。除了扁平文本缓冲区或向量存储，一些方法会显式地构建记忆结构以支持推理。Think-in-

Memory (TiM) stores evolving chains-of-thought to improve consistency across long-horizon reasoning, while A-MEM (Xu et al., 2025) is inspired by Zettelkasten-style linking of notes/experiences. These methods highlight the value of representing intermediate reasoning traces or explicit links, but many retrieval pipelines still predominantly rely on semantic similarity as the primary access mechanism. Graph-based approaches have recently gained traction as a way to capture cross-document and cross-episode dependencies. GraphRAG (Edge et al., 2024b) builds entity-centric graphs and community summaries to answer more global questions over large corpora. Zep proposes a temporally-aware knowledge-graph engine (Graphiti) that synthesizes conversational and structured business data while preserving historical relations. The main text notes these graph-based lines explicitly and motivates a key gap: many systems organize memory around associative proximity (semantic relatedness) rather than mechanistic dependency.

Causal reasoning and long-horizon evaluation. Causal reasoning has been highlighted as both important and challenging for LLMs. The work (Kici-man et al., 2023) study LLMs’ ability to generate causal arguments across multiple causal tasks and emphasize robustness/failure modes, reinforcing that what happened retrieval is not sufficient for why reasoning in many settings. Benchmarking efforts such as LoCoMo (Maharana et al., 2024) stress long-range temporal and causal dynamics in multi-session conversations and provide evaluation tasks that expose long-horizon memory deficits. The paper’s experimental setup also uses Long-MemEval (Wu et al., 2024) as an ultra-long context stress test, and evaluates via LLM-as-a-Judge protocols standard in modern instruction-following evaluation. Overall, prior work demonstrates steady progress in (i) scaling context length, (ii) improving retrieval pipelines, and (iii) building structured, evolving memories for agents. The main text positions MAGMA within this trajectory by explicitly targeting multi-relational structure (semantic/temporal/causal/entity) and intent-aware retrieval control.

B System Implementation Details

B.1 Hyperparameter Configuration

Table 6 presents the comprehensive configuration used in our experiments. These parameters were empirically optimized on the LoCoMo benchmark.

Notably, MAGMA employs an *Adaptive Scoring* mechanism where weights (λ) shift dynamically based on the detected query intent.

Table 6: Hyperparameter settings for MAGMA. "Traversal Weights" correspond to the intent-specific vector $\mathbf{w}_{T_q}$, while λ_1 and λ_2 control the global balance between structural alignment and semantic affinity (Eq. 5).

Module	Parameter	Value/Range
Embedding	Model (Default)	all-MiniLM-L6-v2
	Model (Optional)	text-embedding-3-small
Inference	Dimension	384 / 1536
	LLM Backbone	gpt-4o-mini
Retrieval (Phase 1)	Temperature	0.0
	RRF Constant (k)	60
	Vector Top-K	20
	$w_{keyword}$ (Fusion)	2.0 – 5.0
Traversal (Phase 2)	Sim. Threshold	0.10–0.30
	Max Depth	5 hops
	Max Nodes	200
	Drop Threshold	0.15
Adaptive Weights	λ_1 (Structure Coef.)	1.0 (Base)
	λ_2 (Semantic Coef.)	0.3 – 0.7
	w_{entity} (in $\mathbf{w}_{T_q}$)	2.5 – 6.0
	$w_{temporal}$ (in $\mathbf{w}_{T_q}$)	0.5 – 4.0
	w_{causal} (in $\mathbf{w}_{T_q}$)	3.0 – 5.0
	w_{phrase} (in $\mathbf{w}_{T_q}$)	2.5 – 5.0

C Prompt Library

MAGMA employs a sophisticated prompt strategy with three distinct types, each optimized for specific cognitive tasks within the memory pipeline.

C.1 Event Extraction Prompt (JSON-Structured)

To ensure robustness against hallucination and parsing errors, this module employs a strict JSON schema enforcement strategy. The prompt explicitly defines the extraction targets to ensure downstream graph integrity, capturing not just entities but also semantic relationships and temporal markers.

System Prompt: Event Extractor
System Role: You are an automated Graph Memory Parser. Your task is to extract structured metadata from raw conversational logs to build a knowledge graph. Input Data: - Speaker: {speaker} - Text: {text} - Context: {prev_summary} Instructions: Analyze the input and return ONLY a valid JSON object matching the specific schema below. Do not include markdown formatting. Target Schema: "entities": List of proper nouns (People, Locations, Organizations). "topic": String (1–3 words representing the main theme).

Memory (TiM) 存储不断演化的思维链，以提高长程推理的一致性；而 A-MEM (Xu et al., 2025年) 则受到 Zettelkasten 式笔记/经验链接的启发。这些方法凸显了表示中间推理轨迹或显式链接的价值，但许多检索管道仍然主要依赖语义相似性作为主要的访问机制。基于图的方法最近作为一种捕捉跨文档和跨事件依赖关系的方式而受到关注。GraphRAG (Edgeet al., 2024b) 构建以实体为中心的图和社区摘要，以回答关于大型语料库的更全局性问题。Zep 提出了一个时序感知的知识图谱引擎 (Graphiti)，它综合了对话数据和结构化业务数据，同时保留了历史关系。正文明确指出了这些基于图的研究方向，并揭示了一个关键差距：许多系统围绕关联邻近性（语义相关性）而非机制依赖性来组织记忆。

因果推理与长视野评估。 因果推理已被强调为对大语言模型既重要又具有挑战性。研究（Kici-man et al., 2023）探讨了大语言模型在多种因果任务中生成因果论证的能力，并强调了鲁棒性/失败模式，强化了在许多场景下，仅检索“发生了什么”不足以进行“为什么”的推理。诸如LoCoMo (Maharana et al., 2024) 等基准测试工作强调了多会话对话中的长程时序与因果动态，并提供了暴露长视野记忆缺陷的评估任务。本文的实验设置也使用Long-MemEval (Wu et al., 2024) 作为超长上下文压力测试，并通过现代指令遵循评估中标准的LLM即法官协议进行评估。总体而言，先前的工作展示了在 (i) 扩展上下文长度，(ii) 改进检索管道，以及 (iii) 为智能体构建结构化、演化的记忆方面取得的稳步进展。正文将MAGMA定位于此发展轨迹中，明确针对多关系结构（语义/时序/因果/实体）和意图感知的检索控制。

B 系统实现细节

B.1 超参数配置

表6展示了我们实验中使用的完整配置。这些参数是在LoCoMo基准测试上根据经验优化得出的。

值得注意的是，MAGMA采用了一种自适应评分机制，其中权重 (λ) 会根据检测到的查询意图动态调整。

表6: MAGMA的超参数设置。“遍历权重”对应于意图特定的向量 $\mathbf{w}_{T_q}$ ，而 λ_1 和 λ_2 控制结构对齐与语义亲和度之间的全局平衡（公式5）。

模块	参数	值/范围
嵌入	模型（默认）	all-MiniLM-L6-v2
	模型（可选）	text-embedding-3-small
推理	维度	384 / 1536
	LLM骨干模型	gpt-4o-mini
检索（第一阶段）	温度参数	0.0
	RRF常数 (k)	60
	向量Top-K	20
	$w_{keyword}$ (融合)	2.0 – 5.0
遍历（第二阶段）	相似度阈值	0.10–0.30
	最大深度	5 跳数
	最大节点数	200
	丢弃阈值	0.15
自适应权重	λ_1 (结构系数)	1.0 (基础)
	λ_2 (语义系数)	0.3 – 0.7
	w_{entity} (在 $\mathbf{w}_{T_q}$)	2.5 – 6.0
	$w_{temporal}$ (内宽 T_q)	0.5 – 4.0
	w_{causal} (内宽 T_q)	3.0 – 5.0
	w_{phrase} (内宽 T_q)	2.5 – 5.0

C 提示词库

MAGMA 采用一套精密的提示词策略，包含三种不同的类型，每种都针对记忆流水线中的特定认知任务进行了优化。

C.1 事件提取提示词（JSON结构化）

为确保对幻觉和解析错误的鲁棒性，该模块采用了严格的JSON模式强制执行策略。该提示词明确定义了提取目标，以确保下游图的完整性，不仅捕获实体，还捕获语义关系和时序标记。

系统提示：事件提取器
系统角色： 你是一个自动化图记忆解析器。你的任务是从原始对话日志中提取结构化元数据，以构建一个知识图谱。 输入数据： - 说话者: {speaker} - 文本: {text} - 上下文: {prev_摘要} 指令： 分析输入数据并返回仅一个符合下方特定模式的有效JSON对象。请勿包含Markdown格式化内容。 目标模式： "实体": 专有名词列表（人物、地点、组织）。 "主题": 字符串（1–3个词，代表主要主题）。

- "relationships": List of strings describing interactions (e.g., "X researches Y").
- "semantic_facts": List of atomic facts preserving key information.
- "dates_mentioned": List of temporal strings (e.g., "next Friday", "2024-01-01").
- "summary": One-sentence summary preserving speaker attribution.

C.2 Query-Adaptive QA Prompt

The generation prompt begins with a strict persona definition and appends specific reasoning instructions dynamically based on the Router’s classification (e.g., Multi-hop, Temporal, Open-domain).

System Prompt: Adaptive QA

System Role: You are a precision QA assistant operating on retrieved memory contexts. Your goal is to answer the user’s question accurately using *only* the provided information.
Context: {context}
Current Query:
• Question: {question}
• Constraints: {category_specific_constraints}
Instructions:
1. Use **ONLY** information explicitly stated in the context.
2. If the answer is not present, respond exactly with "Information not found".
3. Be concise (typically 1–10 words) unless detailed reasoning is required.
4. **{dynamic_instruction}** // *Automatically generated by our engine’s query classifier/router (no oracle labels)*
Answer:

*Dynamic Instruction Injection Candidates:

- **[Multi-hop]:** "Connect related facts across different nodes. For comparison queries (e.g., 'both/all'), identify commonalities between entities rather than listing individual details."
- **[Temporal]:** "Resolve relative dates (e.g., 'yesterday') using the event timestamps. Output dates strictly in 'D Month YYYY' format. Calculate durations if asked."
- **[Open-Domain/Inference]:** "Make reasonable inferences based on the user’s personality traits, interests, and past behaviors. Support hypothetical ('would/could') reasoning with evidence."
- **[Single-hop/Factual]:** "Extract the specific entity, name, or method requested. Do not add explanations. Return the ex-

act fact matching the query intent."

C.3 Evaluation Prompt (LLM-as-a-Judge)

To ensure rigorous evaluation beyond simple n-gram overlapping, we employ a semantic scoring mechanism. The Judge LLM evaluates the alignment between the generated response and the ground truth using the following schema.

System Prompt: Semantic Grader

You are an expert evaluator assessing the semantic fidelity of a memory retrieval system. Score the Candidate Answer against the Gold Reference on a continuous scale [0.0, 1.0].
Scoring Rubric:
• **1.0 (Exact Alignment):** Captures all key entities, temporal markers, and causal relationships. Semantically equivalent.
• **0.8 (Substantially Correct):** Main point is accurate but lacks minor nuances or secondary details.
• **0.6 (Partial Match):** Contains valid information but misses key constraints (e.g., wrong date but correct event).
• **0.4 (Tangential):** Touches on the topic but misses the core information requirement.
• **0.2 (Incoherent):** Factually incorrect with only minimal topical overlap.
• **0.0 (Contradiction/Hallucination):** Completely unrelated or contradicts the ground truth.
Evaluation Constraints:
1. **Temporal Flexibility:** Accept relative time references (e.g., "next Tuesday") if they resolve to the same period as the Gold Reference.
2. **Semantic Equivalence:** Prioritize informational content over lexical matching.
3. **Adversarial Handling:** If the Gold Reference states "Unanswerable", the Candidate **MUST** explicitly state lack of information. Any hallucinated fact results in 0.0.
Input: Question: {question} | Gold: {gold} | Candidate: {generated}
Output: JSON {"score": float, "reasoning": "concise explanation"}

D Baseline Configurations

To ensure a fair and rigorous comparison, we standardized the experimental environment across all systems. Specifically, we adhered to the following protocols:

- **Full Context Baseline:** We implemented a "Full Context" baseline where the entire available conversation history is fed directly into the LLM’s context window (up to the 128k token limit of gpt-4o-mini). This serves as a "brute-force" reference to evaluate the model’s native long-context capabilities without external retrieval mechanisms.

- "关系": 描述交互的字符串列表（例如，"X 研究 Y"）。
- "语义_事实": 保留关键信息的原子事实列表。

- "提及_日期": 时序字符串列表（例如，"下周五"、"2024-01-01"）。
- "摘要": 保留说话者归属的单句摘要。

C.2 查询自适应问答提示

生成提示词以严格的人物定义开始，并根据 Router 的分类（例如，多跳、时序、开放域）动态附加特定的推理指令。

系统提示：自适应问答

系统角色：你是一个基于检索到的记忆上下文运行的精准问答助手。你的目标是仅使用提供的信息来准确回答用户的问题。
上下文: {context}
当前查询：
• 问题: {question}
• 约束: {category_specific_constraints}
指令：
1. 仅使用上下文中明确陈述的信息。2. 如果答案不存在，请准确回复“未找到相关信息”。3. 除非需要详细推理，否则请保持简洁（通常为1-10个词）。4. **{dynamic_instruction}** // 由我们引擎的查询分类器/Router 自动生成（无 Oracle 标签）
答案：

*动态指令注入候选：

- **[多跳]:** “连接不同节点间的相关事实。对于比较性查询（例如‘两者/所有’），识别实体间的共性，而非罗列个体细节。”
- **[时序]:** “使用事件时间戳解析相对日期（例如‘昨天’）。输出日期严格采用‘D Month YYYY’格式。若被要求，则计算持续时间。”
- **[开放域/推理]:** “基于用户的个性特征、兴趣和过往行为进行合理推理。用证据支持假设性（‘会/可能’）推理。”

- **[单跳/事实性]:** “提取所请求的具体实体、名称或方法。不要添加解释。返回与

查询意图匹配的确切事实。”

C.3 评估提示词 (LLM即法官)

为确保超越简单的n-gram重叠的严格评估，我们采用了一种语义评分机制。法官LLM使用以下模式评估生成响应与标准答案之间的对齐程度。

系统提示词：语义评分器

你是一位评估记忆检索系统语义保真度的专家评估员。请根据黄金参考，在连续量表 [0.0, 1.0] 上为候选答案评分。
评分标准：
• **1.0 (精确对齐):** 捕捉所有关键实体、时序标记和因果关系。语义上完全等同。
• **0.8 (基本正确):** 主要观点准确，但缺少细微差别或次要细节。
• **0.6 (部分匹配):** 包含有效信息，但遗漏了关键约束条件（例如，日期错误但事件正确）。
• **0.4 (相关但偏离核心):** 触及主题，但未满足核心信息要求。
• **0.2 (不连贯):** 事实错误，仅与主题有最低程度的重叠。
• **0.0 (矛盾/幻觉):** 完全无关或与事实依据相矛盾。

评估约束：
1. **时间灵活性:** 接受相对时间引用（例如“下周二”），只要其解析出的时间段与黄金参考相同。2. **语义等价性:** 优先考虑信息内容而非词汇匹配。3. **对抗性处理:** 如果黄金参考声明“无法回答”，则候选答案必须明确声明缺乏信息。任何编造的事实都会导致得分为0.0。
输入: 问题: {question} | 标准答案: {gold} | 候选答案: {generated}
输出: JSON {"score": float, "reasoning": "简洁的解释"}

D 基线配置

为确保公平且严谨的比较，我们在所有系统中标准化了实验环境。具体而言，我们遵循了以下协议：

- **全上下文基线:** 我们实现了一个“完整上下文”基线模型，将全部可用的对话历史直接输入到大语言模型的上下文窗口中（上限为 gpt-4o-mini 的 128k 令牌限制）。这作为一个“暴力”参考，用于评估模型在不依赖外部检索机制时的原生长上下文处理能力。

• **Retrieval-Based Baselines:** For all baseline systems (e.g., AMem, Nemori, MemoryOS), we applied their official default hyperparameters and storage settings to reflect their standard out-of-the-box performance.

• **Unified Backbone Model:** To eliminate performance variance caused by different foundation models, all systems utilized OpenAI’s gpt-4o-mini for both retrieval reasoning and response generation.

• **Unified Evaluation:** All system outputs were evaluated using the identical *LLM-as-a-Judge* framework (also powered by gpt-4o-mini with temperature=0.0), as detailed in Appendix C.

Dataset Statistics. We conducted a comprehensive evaluation on the full LoCoMo benchmark, testing across all five cognitive categories to assess varying levels of retrieval complexity. The detailed distribution of query types is presented in Table 7.

Table 7: Distribution of query categories in the LoCoMo benchmark used for evaluation.

Query Category	Count
Single-Hop Retrieval	841
Adversarial	446
Temporal Reasoning	321
Multi-Hop Reasoning	282
Open Domain	96
Total Samples	1,986

E Case Study

To demonstrate MAGMA’s reasoning capabilities across different cognitive modalities, we analyze three real-world scenarios from the LoCoMo benchmark. Table 8 provides a side-by-side comparison of MAGMA against key baselines (A-MEM, Nemori, MemoryOS).

E.1 Illustrative Walkthrough: From Memory Construction to Retrieval

To make the case study more concrete, we briefly illustrate how MAGMA processes the Melanie example end to end. Consider three representative facts from the conversation history: (1) Melanie mentions playing the *violin* in an earlier session, (2) she later says that she also plays the *clarinet*, and (3) in a later family-trip session, she refers to

her *son, two children* in a photo, and a hike done *yesterday*.

During memory construction, MAGMA first segments these utterances into event nodes and places them along a temporal backbone. The system then incrementally enriches this memory with additional structure, such as semantic connections between related musical events, entity-centric links for recurring references to Melanie and her family members, and normalized temporal attributes for relative expressions such as *yesterday*. As a result, the memory is not stored as a flat list of text snippets, but as a small multi-view graph in which the same history can be accessed through different relational paths.

At query time, MAGMA first identifies the dominant retrieval intent and then selects anchor events before traversing the most relevant graph views. For example, for “*What instruments does Melanie play?*”, retrieval focuses on the entity and semantic views, allowing MAGMA to aggregate both the earlier *violin* mention and the later *clarinet* mention. For “*How many children does Melanie have?*”, retrieval centers on the local entity neighborhood, combining the references to a *son, two children*, and *brother* into a single evidence set. For “*When did she hike?*”, MAGMA relies primarily on the temporal view, using the normalized representation of *yesterday* to recover the grounded date.

This example highlights the key intuition behind MAGMA: memory construction organizes conversational history into complementary relational views, and query-time retrieval activates different parts of this structure depending on the reasoning need.

E.2 Detailed Analysis

Case 1: Overcoming Information Loss (Recall). For the query regarding instruments, A-MEM failed completely due to its summarization process abstracting away specific details ("violin") from early sessions. Other RAG baselines only retrieved the "clarinet" due to surface-level semantic matching. MAGMA, however, maintains an entity-centric graph structure. Instead of relying on rigid schemas, MAGMA queries the local neighborhood of the [Entity: Melanie] node. This allows it to capture diverse natural language predicates (e.g., "playing my violin", "started clarinet") and aggregate disjoint facts into a comprehensive answer, demonstrating robustness against information loss.

• **基于检索的基线:** 对于所有基线系统（例如，AMem、Nemori、MemoryOS），我们应用了其官方的默认超参数和存储设置，以反映其标准的开箱即用性能。

• **统一骨干模型:** 为消除不同基础模型导致的性能差异，所有系统均采用OpenAI的gpt-4o-mini进行检索、推理和响应生成。

• **统一评估:** 所有系统输出均使用相同的LLM即法官框架（同样由gpt-4o-mini驱动，温度参数设为=0.0）进行评估，详见附录C。

数据集统计. 我们在完整的LoCoMo基准测试上进行了全面评估，测试覆盖所有五个认知类别，以评估不同级别的检索复杂度。查询类型的详细分布见表7。

表7: 用于评估的LoCoMo基准测试中查询类别的分布。

查询类别	数量
单跳检索	841
对抗性时序推理	446
多跳推理	321
开放域	282
总样本数	96
	1,986

E 案例研究

为展示MAGMA在不同认知模态下的推理能力，我们分析了来自LoCoMo基准测试的三个现实场景。表8 提供了MAGMA与关键基线模型（A-MEM、Nemori、MemoryOS）的并列比较。

E.1 示例演练：从记忆构建到检索

为了使案例研究更具体，我们简要说明MAGMA如何端到端处理梅兰妮的示例。考虑对话历史中的三个代表性事实：(1) 梅兰妮在之前的会话中提到拉小提琴，(2) 她后来还说她也演奏单簧管，以及(3) 在之后的家庭旅行会话中，她提到了

她的儿子、照片中的两个孩子，以及昨天完成的一次徒步昨天。

在记忆构建过程中，MAGMA首先将这些话语切分为事件节点，并将它们沿一条时间骨干放置。随后，系统通过附加结构逐步丰富此记忆，例如相关音乐事件之间的语义连接、对梅兰妮及其家庭成员重复提及的以实体为中心的链接，以及对相对表达（如昨天）的规范化时间属性。因此，记忆并非存储为文本片段的扁平列表，而是存储为一个小的多视图图，其中相同的历史可以通过不同的关系路径进行访问。

在查询时间，MAGMA首先识别主导的检索意图，然后选择锚点事件，再遍历最相关的图视图。例如，对于“梅兰妮会演奏什么乐器？”，检索侧重于实体和语义视图，使MAGMA能够聚合较早提及的小提琴和较晚提及的单簧管。对于“梅兰妮有多少个孩子？”，检索集中于局部实体邻域，将提及一个儿子、两个孩子和兄弟的引用合并为一个证据集。对于“她什么时候去徒步了？”，MAGMA主要依赖时间视图，使用昨天的规范化表示来还原具体日期。

这个例子突显了MAGMA背后的核心理念：记忆构建将对话历史组织成互补的关系视图，而查询时检索则根据推理需求激活该结构的不同部分。

E.2 详细分析

案例1: 克服信息丢失（回忆）. 对于关于乐器的查询，A-MEM由于其摘要过程抽象掉了早期会话中的具体细节（“小提琴”）而完全失败。其他RAG基线模型仅因表层语义匹配而检索到“单簧管”。然而，MAGMA维护着一个以实体为中心的图结构。MAGMA不依赖刚性模式，而是查询 [实体: 梅兰妮] 节点的局部邻居。这使得它能够捕获多样化的自然语言谓词（例如，“正在拉我的小提琴”，“开始学习单簧管”），并将离散事实聚合为一个全面答案，展示了对信息丢失的鲁棒性。

Table 8: Case study for failure analysis comparing MAGMA against baselines across three reasoning types. **Red text** indicates hallucinations or partial failures; **Teal text** indicates correct reasoning derived from graph traversal.

	Query & Type	Baseline Failure Mode	MAGMA Graph Reasoning (Success)
Fact	Q1: Fact Retrieval "What instruments does Melanie play?"	A-MEM: "Memories do not explicitly state..." MemoryOS: "Clarinet" Failure: Baselines relying on top-k vector search missed the distant memory of the "violin" (D2:5) because it appeared in a context about "me-time" rather than explicitly about music.	"Clarinet and Violin." Mechanism: MAGMA utilized the entity-centric subgraph around "Melanie". By traversing dynamic semantic edges (e.g., "playing", "enjoy") to connected event nodes, it aggregated all mentions of musical activities regardless of the specific relation label or distance.
	Q2: Logical Inference "How many children does Melanie have?"	Nemori: "At least two..." MemoryOS: "Two" Failure: Baselines performed surface-level extraction from a photo description showing "two children" (D18:5), failing to account for the "son" mentioned in a separate accident event.	"At least three." Mechanism: MAGMA executed multi-hop inference focused on Entity Resolution : 1. Node A (Photo): Identified "two kids" entity. 2. Node B (Accident): Linked "son" (D18:1) via a dynamic relationship edge. 3. Node C (Dialogue): Confirmed "brother" (D18:7) is distinct from the two in the photo. → Logic: 2 (Photo) + 1 (Son/Brother) = 3.
	Q3: Temporal Res. "When did she hike after the roadtrip?"	A-MEM: "20 October 2023" MemoryOS: "29 December 2025" Failure: A-MEM simply copied the session timestamp. MemoryOS hallucinated a future date. Both failed to resolve the relative time expression.	"19 October 2023" Mechanism: MAGMA’s Temporal Parser identified the relative marker "yesterday" in D18:17. Calculation: $T_{\text{session}}(\text{Oct}20) - 1 \text{ day} = \text{Oct}19$. This exact date was anchored to the Event Node, allowing precise retrieval.

Case 2: Multi-Hop Reasoning vs. Surface Extraction. The query "How many children?" exposes a critical weakness in standard RAG: the inability to perform arithmetic across contexts. Baselines simply extracted the explicit mention of "two children" from a photo caption. In contrast, MAGMA treated this as a graph traversal problem focused on entity resolution. It queried the neighborhood of [Entity: Melanie] for connected nodes of type Person. By analyzing the semantic edges, specifically distinguishing the "two kids" entity in the canyon photo from the "son" entity involved in the car accident, MAGMA synthesized these distinct nodes. It correctly deduced that the "son" (referenced later as "brother") was an additional individual, summing up to a count of "at least three," a logical leap impossible for systems relying solely on vector similarity.

Case 3: Temporal Grounding. When asked "When did she hike?", baselines either hallucinated or defaulted to the conversation timestamp (Oct 20). This ignores the semantic meaning of the user’s statement: "we just did it *yesterday*." MAGMA’s structured ingestion pipeline normalizes relative dates during graph construction. The event was stored with the resolved attribute

date="2023-10-19", making the retrieval trivial and exact, completely bypassing the ambiguity that confused the LLM-based baselines.

F Metric Validation Analysis

To validate our choice of using an LLM-based Judge over traditional lexical metrics, we conducted a granular failure analysis on seven representative test cases. Table 10 details the quantitative breakdown.

F.1 Rationale for Semantic Scoring

Our empirical results reveal two critical failure modes where standard metrics (F1, BLEU-1) directly contradict human judgment:

- False Rewards** (The “Hallucination” Problem): Lexical metrics heavily reward incorrect answers that share surface-level tokens.
 - In Case 3, a direct negation (“compatible” vs. “*not* compatible”) yields a remarkably high F1 of 0.857, treating a fatal contradiction as a near-perfect match.
 - In Case 6, substituting the wrong entity (“John” vs. “Sarah”) still achieves **F1**

表8：故障分析案例研究，比较MAGMA与基线模型在三种推理类型上的表现。**红色文本**表示幻觉或部分失败；**青色文本**表示通过图遍历得出的正确推理。

	查询与类型	基线失败模式	MAGMA图推理（成功）
Fact	Q1：事实检索 “什么乐器梅兰妮会演奏什么？”	A-Mem: "记忆并未明确说明..." MemoryOS: "单簧管" 失败: 依赖 top-k 向量检索的基线模型搜索未能找到关于“小提琴”（D2:5）的遥远记忆，因为它出现在一个关于“个人时间”而非明确关于音乐。	“单簧管与小提琴。” 机制：MAGMA 利用了实体-By以“梅兰妮”为中心的图。遍历动态语义边（例如，“playing”、“enjoy”）到连接的事件节点, 它汇总了所有音乐活动的提及, 不论具体关系标签如何或距离。
	Q2：逻辑推理-ence “梅兰妮有多少个孩子？”	Nemori: "至少有两个..." MemoryOS: "两个" 失败: 基线模型进行了表面层次的从照片描述中提取显示"两个孩子"（D18:5），未能考虑到单独事故事件中提到的"儿子"事件。	"至少三个." 机制：MAGMA执行了多跳推理聚焦于 实体解析 ：1. 节点A（照片）：识别出“两个孩子”实体。2. 节点B（事故）：通过一条动态关系边链接了“儿子”（D18:1）。3. 节点C（对话）：确认“兄弟”（D18:7）是与照片中的两人不同的。→ 逻辑：2 (照片) + 1 (儿子/兄弟) = 3。
	Q3: 时间解析 “她是在公路旅行之后何时去徒步的？”	A-MEM: "2023年10月20日" MemoryOS: "2025年12月29日" 失败: A-Mem 只是简单地复制了会话时间戳。MemoryOS 幻觉出了一个未来的日期。两者都未能解析这个相对时间表达式。	"2023年10月19日" 机制：MAGMA 的时间解析器识别出相对时间标记“昨天”在 D18:17 中。计算: $T_{\text{session}}(\text{Oct}20) - 1 \text{ 天} = \text{Oct}19$ 。这个确切日期被锚定到事件节点, 从而实现精确检索。

案例2：多跳推理 vs. 表层提取。 查询“有多少个孩子？”暴露了标准RAG的一个关键弱点：无法跨上下文执行算术运算。基线模型只是从照片标题中提取了“两个孩子”的明确提及。相比之下，MAGMA将此视为一个专注于实体解析的图遍历问题。它查询了 [实体：梅兰妮]的邻居，寻找类型为“人物”的相连节点。通过分析语义边，特别是区分峡谷照片中的“两个孩子”实体与涉及车祸的“儿子”实体，MAGMA综合了这些不同的节点。它正确地推断出“儿子”（后来被提及为“兄弟”）是另一个个体，总计为“至少三个”，这是一个仅依赖向量相似性的系统无法实现的逻辑飞跃。

案例3：时间锚定。 当被问及“她何时去徒步？”时，基线模型要么产生幻觉，要么默认使用对话的时间戳（10月20日）。这忽略了用户陈述的语义含义：“我们昨天刚去过。”MAGMA的结构化摄取流水线在图构建过程中会规范化相对日期。该事件被存储时已解析出属性

日期=“2023-10-19”，使得检索变得简单且精确，完全绕过了那些困扰基于大语言模型的基线模型的歧义。

F 指标验证分析

为了验证我们选择使用基于大语言模型的评判器而非传统词汇指标，我们对七个代表性测试用例进行了细致的故障分析。表10详细列出了定量分析结果。

F.1 语义评分的理由

我们的实证结果揭示了标准指标（F1分数、BLEU-1分数）与人工判断直接矛盾的两种关键失败模式：

- 错误奖励**（“幻觉”问题）：词汇指标会严重奖励那些共享表层标记的错误答案。
 - 在案例3中，一个直接否定（“兼容”与“不兼容”）产生了高达0.857的F1分数，将一个致命矛盾视为近乎完美的匹配。
 - 在案例6中，即使替换了错误的实体（“约翰”与“莎拉”），仍能达到**F1分数**

Table 9: LoCoMo evaluation with F1 and BLEU-1 metrics

Method	Multi-Hop		Temporal		Open-Domain		Single-Hop		Overall	
	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1	F1	BLEU-1
Full Context	0.182	0.128	0.079	0.055	0.042	0.030	0.229	0.156	0.140	0.096
A-MEM	0.128	0.088	0.128	0.079	0.076	0.051	0.174	0.110	0.116	0.074
MemoryOS	0.365	0.276	0.434	0.369	0.246	0.191	0.493	0.437	0.413	0.355
Nemori	0.363	0.249	0.569	0.479	0.247	0.189	0.548	0.439	0.502	0.403
MAGMA (ours)	0.264	0.172	0.509	0.370	0.180	0.136	0.551	0.477	0.467	0.378

0.750, rewarding the hallucinatory output.

2. **False Penalties** (The “Phrasing” Problem): Valid answers with different formatting or synonyms are unfairly penalized.

- In Case 4 (Time Notation) and Case 5 (Synonyms), F1 and BLEU scores drop to 0.000 despite the answers being semantically identical.

As shown in Table 10, the LLM-Judge correctly assigns a score of 0.0 to factual errors and 1.0 to semantic matches, aligning perfectly with reasoning requirements.

表 9：使用F1分数和BLEU-1分数指标进行的LoCoMo评估

方法	多跳		时序		开放域		单跳		总体	
	F1	BLEU-1分数	F1	BLEU-1分数	F1	BLEU-1分数	F1	BLEU-1分数	F1	BLEU-1分数
完整上下文	0.182	0.128	0.079	0.055	0.042	0.030	0.229	0.156	0.140	0.096
A-Mem	0.128	0.088	0.128	0.079	0.076	0.051	0.174	0.110	0.116	0.074
MemoryOS	0.365	0.276	0.434	0.369	0.246	0.191	0.493	0.437	0.413	0.355
Nemori	0.363	0.249	0.569	0.479	0.247	0.189	0.548	0.439	0.502	0.403
MAGMA（我们的）	0.264	0.172	0.509	0.370	0.180	0.136	0.551	0.477	0.467	0.378

0.750，这反而奖励了这种幻觉输出。

2. **错误惩罚**（“措辞”问题）：因格式化或同义词不同而导致的有效答案被不公正地惩罚。

- 在案例4（时间表示法）和案例5（同义词）中，尽管答案在语义上完全相同，但F1分数和Bleu分数却降至0.000。

如表10所示，LLM评判器能正确地事实错误分配0.0分，为语义匹配分配1.0分，这与推理要求完美契合。

Table 10: Quantitative Failure Analysis of Lexical Metrics. We present seven controlled cases with their calculated F1 and BLEU-1 scores. The data demonstrates that lexical metrics frequently assign high scores to fatal errors (False Rewards) and zero scores to correct variations (False Penalties), whereas the LLM-Judge correctly assesses semantic validity.

Failure Mode	Case Detail (Gold / Predicted)	Lexical Metrics (F1 / BLEU-1)	LLM Judge (Semantic)
Case 1: False Reward (Wrong Fact, High Overlap)	Gold: “three items” Pred: “five items” <i>Analysis:</i> Factually wrong count, but rewarded for sharing the noun “items”.	High 0.500 / 0.500	0.0 (Reject)
Case 2: False Penalty (Verbose Phrasing)	Gold: “18 days” Pred: “The total duration was 18 days” <i>Analysis:</i> Correct answer penalized for low precision due to extra words.	Low 0.500 / 0.333	1.0 (Accept)
Case 3: False Reward (Negation/Contradiction)	Gold: “compatible with Mac” Pred: “ not compatible with Mac” <i>Analysis:</i> Fatal contradiction receives near-perfect scores due to high token overlap.	Very High 0.857 / 0.750	0.0 (Reject)
Case 4: False Penalty (Time Notation)	Gold: “14:00” Pred: “2 PM” <i>Analysis:</i> Different formats result in zero overlap despite identical meaning.	Zero 0.000 / 0.000	1.0 (Accept)
Case 5: False Penalty (Synonyms)	Gold: “cheap” Pred: “inexpensive” <i>Analysis:</i> Standard metrics cannot handle synonym matching without external resources.	Zero 0.000 / 0.000	1.0 (Accept)
Case 6: False Reward (Entity Hallucination)	Gold: “John completed the project” Pred: “Sarah completed the project” <i>Analysis:</i> Wrong entity (Sarah vs John), yet high metrics due to shared sentence structure.	High 0.750 / 0.750	0.0 (Reject)
Case 7: False Penalty (Format Noise)	Gold: “5” Pred: “5 (extracted from JSON...)” <i>Analysis:</i> Correct value embedded in noise results in poor precision metrics.	Low 0.286 / 0.167	1.0 (Accept)

表10：词汇指标的定量失败分析。我们展示了七个受控案例及其计算出的F1分数和BLEU-1分数。数据表明，词汇指标经常为致命错误（错误奖励）分配高分，而为正确的变体（错误惩罚）分配零分，而LLM评判器则能正确评估语义有效性。

失败模式	案例详情（标准答案 / 预测答案）	词汇指标 rics (F1分数 / BLEU-1分数)	LLM评判器 (语义)
案例 1：虚假奖励 (事实错误，高重叠)	标准答案：“three items” 预测答案：“five items” 分析：事实计数错误，但仍获得奖励用于共享名词“items”。	High 0.500 / 0.500	0.0 (拒绝)
案例2: 误罚 (冗长措辞)	标准答案：“18 days” 预测答案: “总时长为18天” 分析：正确答案因精确度低（由于额外词汇）而被扣分。决策因多余词汇而受影响。	Low 0.500 / 0.333	1.0 (接受)
案例3：虚假奖励 (否定/矛盾)	标准答案: “兼容Mac” 预测答案: “不兼容Mac” 分析：致命矛盾获得接近完美的分数。	非常高 0.857 / 0.750	0.0 (拒绝)
案例4: 误罚 (时间表示法)	标准答案：“14:00” 预测答案：“下午2点” 分析：不同的格式化导致零重叠，尽管含义完全相同。	Zero 0.000 / 0.000	1.0 (接受)
案例5: 误罚 (同义词)	标准答案：“便宜的” 预测答案: “不贵的” 分析：标准指标无法处理同义词匹配，除非借助外部资源。	Zero 0.000 / 0.000	1.0 (接受)
案例6: 虚假奖励 (实体幻觉)	标准答案: “约翰完成了项目” 预测答案: “莎拉完成了项目” 分析：错误实体（莎拉 vs 约翰），然而因共享句子结构而导致指标偏高。	High 0.750 / 0.750	0.0 (拒绝)
案例7: 误罚 (格式噪声)	标准答案：“5” 预测答案: “5（从JSON中提取...）” 分析：正确值被嵌入在噪声中，导致精确度指标不佳。	Low 0.286 / 0.167	1.0 (接受)